

Radiation From Moving Charges

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Abstract

Yet again, the questions which were bothering physicists at the beginning of the 20th century, and which I believe still trouble us even 100 years later (see [1], [2], [3]), have now begun to trouble me as well.

Will a freely falling charge radiate electromagnetic radiation?

Will electromagnetic radiation emitted by a point charge moving in the field of a stationary point charge cause the moving charge to spiral toward the stationary charge?

If so, what about the electron in the Rutherford atomic model? Will it fall onto the nucleus?

What is the source of the radiation?

Will constant current flowing through a toroidal (ring-shaped) wire produce electromagnetic waves?

What about the Earth orbiting the Sun? Why does it not lose energy and spiral toward the Sun? Or perhaps it does, but very, very slowly?

Despite everything, we still have the same inverse-square law governing electrostatics and gravitation.

This article is an attempt to answer these questions. It will proceed in a rather detailed way. Step by step. Starting from the general solutions of Maxwell's equations, through the Liénard–Wiechert potentials, to the radiation of a point charge moving first with uniform and then with arbitrary velocity. Once we obtain the formulas for the fields of an arbitrarily moving point charge, we will discuss several commonly encountered cases.

So, let us begin.

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1 Maxwell's equations

This is the beginning of our journey. Four Maxwell's equations that are the basis of all our further derivations. I will present them in differential form.

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} \quad (1.1)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (1.2)$$

$$\nabla \cdot \mathbf{B} = 0 \quad (1.3)$$

$$\nabla \times \mathbf{B} = \frac{\mathbf{j}}{\epsilon_0 c^2} + \frac{1}{c^2} \frac{\partial \mathbf{E}}{\partial t} \quad (1.4)$$

2 From Maxwell's to wave equations

Each of the Maxwell's equations is a partial differential equation (PDE) of 1st order. It appears that this set of the 4 Maxwell's equations can be transformed to the equivalent set of two 2nd order PDEs. Let me provide the derivations.

We begin with the equation $\nabla \cdot \mathbf{B} = 0$, known as Gauss's law for magnetism. To solve that equation (to find $\mathbf{B}$) we will use one of the vector calculus identities, the one saying that divergence of the curl of any vector field is equal to zero

$$\nabla \cdot (\nabla \times \mathbf{A}) = 0. \quad (2.1)$$

So now, using that identity we may write

$$\mathbf{B} = \nabla \times \mathbf{A} \quad (2.2)$$

where $\mathbf{A}$ is some arbitrary vector field, which we now start calling vector potential. So, we found $\mathbf{B}$. But (if you have noticed) we are cheating a bit here. This is because at the same time we have introduced a new variable, a vector potential $\mathbf{A}$, which now needs to be found somehow. But that's ok. We will deal with that soon.

Let us try now to do similar thing for $\mathbf{E}$. To do so, we will start from Faraday's law, $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$.

If we now substitute $\nabla \times \mathbf{A}$ for $\mathbf{B}$, we get

$$\nabla \times \mathbf{E} = -\frac{\partial}{\partial t} \nabla \times \mathbf{A}. \quad (2.3)$$

And because we can differentiate either with respect to time or to space first, we can also write above equation as

$$\nabla \times \mathbf{E} = -\nabla \times \frac{\partial \mathbf{A}}{\partial t}, \quad (2.4)$$

or as

$$\nabla \times \left(\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t} \right) = 0. \quad (2.5)$$

Now, using another vector calculus identity, the one saying that the curl of the gradient of any scalar field is equal to zero

$$\nabla \times (\nabla \phi) = 0, \quad (2.6)$$

we may write that the vector field $\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t}$ shall be equal to the gradient of some arbitrary scalar function ϕ

$$-\nabla \phi = \mathbf{E} + \frac{\partial \mathbf{A}}{\partial t}, \quad (2.7)$$

which we now start calling a scalar potential. Minus sign at the front of the $\nabla \phi$ has been added for some "compatibility" reasons.

We may now rewrite last equation as

$$\mathbf{E} = -\nabla \phi - \frac{\partial \mathbf{A}}{\partial t}, \quad (2.8)$$

and just because of that minus sign, the equation (2.8) will now be compliant with the well known electrostatic variant $\mathbf{E} = -\nabla \phi$ where $\frac{\partial \mathbf{A}}{\partial t}$ disappears as in that case nothing changes in time.

So, we have found $\mathbf{B}$ and $\mathbf{E}$, but at the same time we have introduced two new variables, vector and scalar potentials, $\mathbf{A}$ and ϕ respectively. Let's try then to find relations for those potentials. At the end, we still have two Maxwell's equations which we haven't used yet.

Let's start from the Gauss' law $\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$. If we substitute $-\nabla \phi - \frac{\partial \mathbf{A}}{\partial t}$ for $\mathbf{E}$, (see (2.8)) we will get

$$\nabla \cdot \left(-\nabla \phi - \frac{\partial \mathbf{A}}{\partial t} \right) = \frac{\rho}{\epsilon_0}, \quad (2.9)$$

which after multiplication can be written as

$$-\nabla^2\phi - \frac{\partial}{\partial t}\nabla \cdot \mathbf{A} = \frac{\rho}{\epsilon_0}. \quad (2.10)$$

If we do the same with the 4th of the Maxwell's equations, $\nabla \times \mathbf{B} = \frac{\mathbf{j}}{\epsilon_0 c^2} + \frac{1}{c^2} \frac{\partial \mathbf{E}}{\partial t}$ (the Ampere's law), I mean if we substitute $-\nabla\phi - \frac{\partial \mathbf{A}}{\partial t}$ for $\mathbf{E}$ (see (2.8)) and $\nabla \times \mathbf{A}$ for $\mathbf{B}$ (see (2.2)), we will get

$$\nabla \times (\nabla \times \mathbf{A}) = \frac{\mathbf{j}}{\epsilon_0 c^2} + \frac{1}{c^2} \frac{\partial}{\partial t} \left(-\nabla\phi - \frac{\partial \mathbf{A}}{\partial t} \right). \quad (2.11)$$

At this moment we will use yet another vector calculus identity

$$\nabla \times (\nabla \times \mathbf{A}) = \nabla (\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}. \quad (2.12)$$

Thus equation (2.11) can be written as

$$\nabla (\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} + \frac{1}{c^2} \frac{\partial}{\partial t} \nabla\phi + \frac{1}{c^2} \frac{\partial^2 \mathbf{A}}{\partial t^2} = \frac{\mathbf{j}}{\epsilon_0 c^2}, \quad (2.13)$$

or after small rearrangement as

$$\nabla^2 \mathbf{A} - \frac{1}{c^2} \frac{\partial^2 \mathbf{A}}{\partial t^2} - \nabla \left(\nabla \cdot \mathbf{A} + \frac{1}{c^2} \frac{\partial \phi}{\partial t} \right) = -\frac{\mathbf{j}}{\epsilon_0 c^2}. \quad (2.14)$$

At that moment we have reached our goal. We've transformed 4 Maxwell's equations into the two equivalent 2nd order PDEs of vector $\mathbf{A}$ and scalar ϕ potentials. These are (2.10) and (2.14). But equations (2.10) and (2.14) are coupled. This is not very convenient. It would be great if we could decouple them. It appears that we can do so. To do this, let us notice that magnetic field $\mathbf{B} = \nabla \times \mathbf{A}$ will not change if instead of $\mathbf{A}$ we will use $\mathbf{A}' = \mathbf{A} + \nabla\Lambda$. Λ is here an arbitrary scalar potential, and because the curl of $\nabla\Lambda$ is always zero, $\mathbf{B}$ will not change by such transformation. If at the same time we augment scalar potential ϕ , so that $\phi' = \phi - \frac{\partial \Lambda}{\partial t}$, the electric field $\mathbf{E} = -\nabla\phi - \frac{\partial \mathbf{A}}{\partial t}$ will not change as well. The potentials are somehow at our disposal. We may choose them arbitrarily as long as the associated fields are not altered. If we could pick Λ , so that relation

$$\nabla \cdot \mathbf{A}' = -\frac{1}{c^2} \frac{\partial \phi'}{\partial t} \quad (2.15)$$

between vector and scalar potentials is true, then our coupled equations (2.10) and (2.14) will gain beautiful, the same form of inhomogeneous wave equation

$$\nabla^2 \phi - \frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2} = -\frac{\rho}{\epsilon_0}, \quad (2.16)$$

$$\nabla^2 \mathbf{A} - \frac{1}{c^2} \frac{\partial^2 \mathbf{A}}{\partial t^2} = -\frac{\mathbf{j}}{\epsilon_0 c^2}. \quad (2.17)$$

And it appears that we can find such a Λ . If we substitute primed potentials $\phi' = \phi - \frac{\partial \Lambda}{\partial t}$ and $\mathbf{A}' = \mathbf{A} + \nabla \Lambda$ into a Lorentz gauge condition (2.15), we obtain the wave equation for the gauge function Λ

$$\nabla^2 \Lambda - \frac{1}{c^2} \frac{\partial^2 \Lambda}{\partial t^2} = -\left(\nabla \cdot \mathbf{A} + \frac{1}{c^2} \frac{\partial \phi}{\partial t} \right). \quad (2.18)$$

That is, the Lorentz gauge function Λ always obeys an inhomogeneous wave equation.

Equations (2.16) and (2.17) look nearly the same. In fact they constitute a set of four equations (one for ϕ and 3 for each component of $\mathbf{A}$) which are all of the same form of the inhomogeneous three-dimensional wave equation

$$\nabla^2 \xi(\mathbf{r}, t) - \frac{1}{c^2} \frac{\partial^2 \xi(\mathbf{r}, t)}{\partial t^2} = -f(\mathbf{r}, t). \quad (2.19)$$

$\xi(\mathbf{r}, t)$ is an unknown potential, and $f(\mathbf{r}, t)$ is some known so-called forcing/source function. The process of finding a solution to this equation is purely mathematical and is covered in section 10.1 "Solution of the inhomogeneous wave equation". Here I am presenting only the final result

$$\xi(\mathbf{r}, t) = \frac{1}{4\pi} \iiint_{V'} \frac{f\left(\mathbf{r}', t - \frac{|\mathbf{r} - \mathbf{r}'|}{c}\right)}{|\mathbf{r} - \mathbf{r}'|} dV'. \quad (2.20)$$

Thus our wave equations (2.16) and (2.17) have their solutions of the following form

$$\phi(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \iiint_{V'} \frac{\rho\left(\mathbf{r}', t - \frac{|\mathbf{r} - \mathbf{r}'|}{c}\right)}{|\mathbf{r} - \mathbf{r}'|} dV', \quad (2.21)$$

$$\mathbf{A}(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0 c^2} \iiint_{V'} \frac{\mathbf{j}\left(\mathbf{r}', t - \frac{|\mathbf{r} - \mathbf{r}'|}{c}\right)}{|\mathbf{r} - \mathbf{r}'|} dV'. \quad (2.22)$$

3 Liénard–Wiechert potentials

Equations (2.21) and (2.22) allow us to find potentials and then the electric and magnetic fields at the field point $(\mathbf{r}, t)$, taking into account the contributions of charges and their currents from all points $(\mathbf{r}', t')$. t' is the so-called retarded time (a time when the contribution was at location $\mathbf{r}'$), and it differs from the time of observation by the following formula

$$t' = t - \frac{|\mathbf{r} - \mathbf{r}'|}{c}. \quad (3.1)$$

This is because a time period $\Delta t = \frac{|\mathbf{r} - \mathbf{r}'|}{c}$ is needed for a contribution to travel with the speed of light c from $\mathbf{r}'$ to $\mathbf{r}$.

Let us now try to find potentials not from the many contributions, but only coming from one, moving with arbitrary velocity point charge. For that let me introduce two vectors. $\boldsymbol{\zeta}(t)$ is the point charge position vector, and the $\mathbf{v}(t) = \frac{d\boldsymbol{\zeta}(t)}{dt}$ is that charge velocity. Then our source functions $\rho(\mathbf{r}, t)$ and $\mathbf{j}(\mathbf{r}, t)$ can be expressed as

$$\rho(\mathbf{r}, t) = q\delta^3(\mathbf{r} - \boldsymbol{\zeta}(t)), \quad (3.2)$$

$$\mathbf{j}(\mathbf{r}, t) = q\delta^3(\mathbf{r} - \boldsymbol{\zeta}(t))\mathbf{v}(t), \quad (3.3)$$

and the potentials (2.21) and (2.22) gain the following shape

$$\phi(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0} \iiint_{V'} \frac{\delta^3\left(\mathbf{r}' - \boldsymbol{\zeta}\left(t - \frac{|\mathbf{r} - \mathbf{r}'|}{c}\right)\right)}{|\mathbf{r} - \mathbf{r}'|} dV', \quad (3.4)$$

$$\mathbf{A}(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0 c^2} \iiint_{V'} \frac{\delta^3\left(\mathbf{r}' - \boldsymbol{\zeta}\left(t - \frac{|\mathbf{r} - \mathbf{r}'|}{c}\right)\right) \mathbf{v}\left(t - \frac{|\mathbf{r} - \mathbf{r}'|}{c}\right)}{|\mathbf{r} - \mathbf{r}'|} dV'. \quad (3.5)$$

If we look at these equations, we will notice that they have a similar form.

$$\phi(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0} \iiint_{V'} \delta^3(\mathbf{f}(\mathbf{r}')) g(\mathbf{r}') dV', \quad (3.6)$$

$$\mathbf{A}(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0 c^2} \iiint_{V'} \delta^3(\mathbf{f}(\mathbf{r}')) \mathbf{h}(\mathbf{r}') dV', \quad (3.7)$$

where

$$\mathbf{f}(\mathbf{r}') = \mathbf{r}' - \boldsymbol{\zeta} \left(t - \frac{|\mathbf{r} - \mathbf{r}'|}{c} \right) \quad (3.8)$$

is the vector function of the vector variable $\mathbf{r}'$,

$$g(\mathbf{r}') = \frac{1}{|\mathbf{r} - \mathbf{r}'|} \quad (3.9)$$

is the scalar function of the vector variable $\mathbf{r}'$, and

$$\mathbf{h}(\mathbf{r}') = \frac{\mathbf{v} \left(t - \frac{|\mathbf{r} - \mathbf{r}'|}{c} \right)}{|\mathbf{r} - \mathbf{r}'|} \quad (3.10)$$

is the vector function of the vector variable $\mathbf{r}'$.

If the argument of the δ function were $\mathbf{r}'$ itself, then the integrals (3.6) and (3.7) would be trivial to evaluate. Straight from the definition of the Dirac delta function we would get

$$\phi(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0} \iiint_{V'} \delta^3(\mathbf{r}') g(\mathbf{r}') dV' = \frac{q}{4\pi\epsilon_0} g(\mathbf{0}), \quad (3.11)$$

$$\mathbf{A}(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0 c^2} \iiint_{V'} \delta^3(\mathbf{r}') \mathbf{h}(\mathbf{r}') dV' = \frac{q}{4\pi\epsilon_0 c^2} \mathbf{h}(\mathbf{0}). \quad (3.12)$$

But we aren't so lucky.

Thus, if the $\mathbf{r}'$ is not the direct argument of the δ function, let us introduce one, let's say $\boldsymbol{\eta}$, which is. Thus

$$\boldsymbol{\eta} = \mathbf{f}(\mathbf{r}'), \quad (3.13)$$

and

$$\mathbf{r}' = \mathbf{f}^{-1}(\boldsymbol{\eta}). \quad (3.14)$$

We must check/ensure that the $\mathbf{f}$ is a bijective function, so that it is invertible. A vector function is invertible if and only if the determinant of its Jacobi matrix is non-zero. So that, we must ensure that

$$\det [\mathbf{J}(\mathbf{f})] = \frac{\partial (\eta_1, \eta_2, \eta_3)}{\partial (r'_1, r'_2, r'_3)} = \begin{vmatrix} \frac{\partial \eta_1}{\partial r'_1} & \frac{\partial \eta_1}{\partial r'_2} & \frac{\partial \eta_1}{\partial r'_3} \\ \frac{\partial \eta_2}{\partial r'_1} & \frac{\partial \eta_2}{\partial r'_2} & \frac{\partial \eta_2}{\partial r'_3} \\ \frac{\partial \eta_3}{\partial r'_1} & \frac{\partial \eta_3}{\partial r'_2} & \frac{\partial \eta_3}{\partial r'_3} \end{vmatrix} \neq 0, \quad (3.15)$$

or, which is equivalent,

$$\det [\mathbf{J}(\mathbf{f}^{-1})] = \frac{\partial (r'_1, r'_2, r'_3)}{\partial (\eta_1, \eta_2, \eta_3)} = \begin{vmatrix} \frac{\partial r'_1}{\partial \eta_1} & \frac{\partial r'_1}{\partial \eta_2} & \frac{\partial r'_1}{\partial \eta_3} \\ \frac{\partial r'_2}{\partial \eta_1} & \frac{\partial r'_2}{\partial \eta_2} & \frac{\partial r'_2}{\partial \eta_3} \\ \frac{\partial r'_3}{\partial \eta_1} & \frac{\partial r'_3}{\partial \eta_2} & \frac{\partial r'_3}{\partial \eta_3} \end{vmatrix} \neq 0. \quad (3.16)$$

Once we are sure that

$$\det [\mathbf{J}(\mathbf{f})] = \frac{1}{\det [\mathbf{J}(\mathbf{f}^{-1})]} \neq 0, \quad (3.17)$$

we may find relationship between infinitesimal volume elements $d^3V' = d^3r'$ and $d^3\eta$. This is

$$d^3r' = \frac{\partial (r'_1, r'_2, r'_3)}{\partial (\eta_1, \eta_2, \eta_3)} d^3\eta. \quad (3.18)$$

With all that information we may now proceed to a change of the vector variable from $\mathbf{r}'$ to $\boldsymbol{\eta} = \mathbf{f}(\mathbf{r}')$ in equations (3.6) and (3.7). After this change of variables, equations (3.6) and (3.7) gain the following form

$$\begin{aligned} \phi(\mathbf{r}, t) &= \frac{q}{4\pi\epsilon_0} \iiint_{V'} \delta^3(\mathbf{f}(\mathbf{r}')) g(\mathbf{r}') dV' = \\ &= \frac{q}{4\pi\epsilon_0} \iiint_{\eta} \delta^3(\boldsymbol{\eta}) g(\mathbf{f}^{-1}(\boldsymbol{\eta})) \frac{\partial (r'_1, r'_2, r'_3)}{\partial (\eta_1, \eta_2, \eta_3)} d^3\eta = \\ &= \frac{q}{4\pi\epsilon_0} g(\mathbf{f}^{-1}(\mathbf{0})) \left. \frac{\partial (r'_1, r'_2, r'_3)}{\partial (\eta_1, \eta_2, \eta_3)} \right|_{\boldsymbol{\eta}=\mathbf{0}} = \\ &= \frac{q}{4\pi\epsilon_0} \frac{1}{|\mathbf{r} - \mathbf{f}^{-1}(\mathbf{0})|} \left. \frac{1}{\frac{\partial (\eta_1, \eta_2, \eta_3)}{\partial (r'_1, r'_2, r'_3)}} \right|_{\boldsymbol{\eta}=\mathbf{0}}, \quad (3.19) \end{aligned}$$

$$\begin{aligned}
\mathbf{A}(\mathbf{r}, t) &= \frac{q}{4\pi\epsilon_0 c^2} \iiint_{V'} \delta^3(\mathbf{f}(\mathbf{r}')) \mathbf{h}(\mathbf{r}') dV' = \\
&= \frac{q}{4\pi\epsilon_0 c^2} \iiint_{\boldsymbol{\eta}} \delta^3(\boldsymbol{\eta}) \mathbf{h}(\mathbf{f}^{-1}(\boldsymbol{\eta})) \frac{\partial(r'_1, r'_2, r'_3)}{\partial(\eta_1, \eta_2, \eta_3)} d^3\eta = \\
&= \frac{q}{4\pi\epsilon_0 c^2} \mathbf{h}(\mathbf{f}^{-1}(\mathbf{0})) \frac{\partial(r'_1, r'_2, r'_3)}{\partial(\eta_1, \eta_2, \eta_3)} \Big|_{\boldsymbol{\eta}=\mathbf{0}} = \\
&= \frac{q}{4\pi\epsilon_0 c^2} \frac{\mathbf{v}\left(t - \frac{|\mathbf{r} - \mathbf{f}^{-1}(\mathbf{0})|}{c}\right)}{|\mathbf{r} - \mathbf{f}^{-1}(\mathbf{0})|} \frac{1}{\frac{\partial(\eta_1, \eta_2, \eta_3)}{\partial(r'_1, r'_2, r'_3)} \Big|_{\boldsymbol{\eta}=\mathbf{0}}}. \quad (3.20)
\end{aligned}$$

The equations (3.19) and (3.20) look a bit enigmatic. What is this $\mathbf{f}^{-1}(\mathbf{0})$ and what is $\frac{\partial(\eta_1, \eta_2, \eta_3)}{\partial(r'_1, r'_2, r'_3)} \Big|_{\boldsymbol{\eta}=\mathbf{0}}$. We will try to figure that out in the following lines.

Let's start with $\mathbf{f}^{-1}(\mathbf{0})$. $\mathbf{r}' = \mathbf{f}^{-1}(\mathbf{0})$ is in fact a solution (with respect to $\mathbf{r}'$) of the equation $\boldsymbol{\eta} = \mathbf{0}$. This is

$$\boldsymbol{\eta} = \mathbf{f}(\mathbf{r}') = \mathbf{r}' - \boldsymbol{\zeta}\left(t - \frac{|\mathbf{r} - \mathbf{r}'|}{c}\right) = \mathbf{0}. \quad (3.21)$$

Thus

$$\mathbf{r}' = \boldsymbol{\zeta}(t'), \quad (3.22)$$

where

$$t' = t - \frac{|\mathbf{r} - \mathbf{r}'|}{c} \quad (3.23)$$

is the so-called retarded time (a time when the charge was on its trajectory at the moment of the emission of the EM wave, which now, at time t was received at position $\mathbf{r}$). Retarded time is also discussed at the beginning of this chapter. So finally, for $\mathbf{f}^{-1}(\mathbf{0})$ we have

$$\mathbf{f}^{-1}(\mathbf{0}) = \boldsymbol{\zeta}(t'). \quad (3.24)$$

Now, let's come back to expression $\frac{\partial(\eta_1, \eta_2, \eta_3)}{\partial(r'_1, r'_2, r'_3)} \Big|_{\boldsymbol{\eta}=\mathbf{0}}$. This is the Jacobian of the vector function $\boldsymbol{\eta} = \mathbf{f}(\mathbf{r}') = \mathbf{r}' - \boldsymbol{\zeta}(t')$. Thus

$$\left. \frac{\partial (\eta_1, \eta_2, \eta_3)}{\partial (r'_1, r'_2, r'_3)} \right|_{\boldsymbol{\eta}=\mathbf{0}} = \begin{vmatrix} \frac{\partial \eta_1}{\partial r'_1} & \frac{\partial \eta_1}{\partial r'_2} & \frac{\partial \eta_1}{\partial r'_3} \\ \frac{\partial \eta_2}{\partial r'_1} & \frac{\partial \eta_2}{\partial r'_2} & \frac{\partial \eta_2}{\partial r'_3} \\ \frac{\partial \eta_3}{\partial r'_1} & \frac{\partial \eta_3}{\partial r'_2} & \frac{\partial \eta_3}{\partial r'_3} \end{vmatrix}. \quad (3.25)$$

To evaluate the Jacobian, we need to perform this tedious work, and calculate each of the 9 differentials $\frac{\partial \eta_i}{\partial r'_j}$ of the determinant.

Let's start this process by expressing the $\boldsymbol{\eta}$ function in the "component" form

$$\eta_i = r'_i - \zeta_i(t') = r'_i - \zeta_i \left(t - \frac{\sqrt{(r_1 - r'_1)^2 + (r_2 - r'_2)^2 + (r_3 - r'_3)^2}}{c} \right) \quad (3.26)$$

Then

$$\frac{\partial \eta_i}{\partial r'_j} = \frac{\partial r'_i}{\partial r'_j} - \frac{\partial \zeta_i(t')}{\partial r'_j} = \delta_{ij} - \frac{\partial \zeta_i(t')}{\partial t'} \frac{\partial t'}{\partial r'_j} \quad (3.27)$$

where $\frac{\partial \zeta_i(t')}{\partial t'} = v_i$ is the i^{th} component of the charge velocity $\mathbf{v}(t')$ at the retarded time t' , and (from equation (3.1))

$$\frac{\partial t'}{\partial r'_j} = \frac{1}{c} \frac{r_j - r'_j}{\sqrt{(r_1 - r'_1)^2 + (r_2 - r'_2)^2 + (r_3 - r'_3)^2}} = \frac{1}{c} \frac{r_j - r'_j}{|\mathbf{r} - \mathbf{r}'|}. \quad (3.28)$$

Vector $\mathbf{r} - \mathbf{r}'$ pops up so often in our derivations so it would be appropriate to assign it a separate symbol. Let's choose $\mathbf{r}^*$ symbol for that purpose. The

$$\mathbf{r}^* = \mathbf{r} - \mathbf{r}' \quad (3.29)$$

is thus the vector from the charge position $\mathbf{r}'$ at the retarded time t' to the point of observation $\mathbf{r}$. Then

$$\mathbf{e}_{r^*} = \frac{\mathbf{r}^*}{|\mathbf{r}^*|} \quad (3.30)$$

is the unit vector which points from the retarded position $\mathbf{r}'$ to the point of observation $\mathbf{r}$, and of course

$$\mathbf{r}^* = r^* \mathbf{e}_{r^*}. \quad (3.31)$$

With all that conventions, we may write

$$\frac{\partial t'}{\partial r'_j} = \frac{e_{r_j^*}}{c} \quad (3.32)$$

where $e_{r_j^*}$ is the j^{th} component of the unit vector $\mathbf{e}_{r^*}$, and finally

$$\frac{\partial \eta_i}{\partial r'_j} = \delta_{ij} - \frac{\partial \zeta_i(t')}{\partial t'} \frac{\partial t'}{\partial r'_j} = \delta_{ij} - \frac{v_i(t')}{c} e_{r_j^*}. \quad (3.33)$$

Thus

$$\left. \frac{\partial (\eta_1, \eta_2, \eta_3)}{\partial (r'_1, r'_2, r'_3)} \right|_{\boldsymbol{\eta}=\mathbf{0}} = \begin{vmatrix} 1 - \frac{v_1(t')}{c} e_{r_1^*} & -\frac{v_1(t')}{c} e_{r_2^*} & -\frac{v_1(t')}{c} e_{r_3^*} \\ -\frac{v_2(t')}{c} e_{r_1^*} & 1 - \frac{v_2(t')}{c} e_{r_2^*} & -\frac{v_2(t')}{c} e_{r_3^*} \\ -\frac{v_3(t')}{c} e_{r_1^*} & -\frac{v_3(t')}{c} e_{r_2^*} & 1 - \frac{v_3(t')}{c} e_{r_3^*} \end{vmatrix}, \quad (3.34)$$

and after rather arduous calculations we get

$$\left. \frac{\partial (\eta_1, \eta_2, \eta_3)}{\partial (r'_1, r'_2, r'_3)} \right|_{\boldsymbol{\eta}=\mathbf{0}} = 1 - \left(\frac{v_1(t') e_{r_1^*} + v_2(t') e_{r_2^*} + v_3(t') e_{r_3^*}}{c} \right) = 1 - \frac{\mathbf{v}(t')}{c} \cdot \mathbf{e}_{r^*}. \quad (3.35)$$

Jacobian $\left. \frac{\partial (\eta_1, \eta_2, \eta_3)}{\partial (r'_1, r'_2, r'_3)} \right|_{\boldsymbol{\eta}=\mathbf{0}}$ will appear so many times in the following derivations that it is convenient to introduce a new symbol for it. Let us choose κ for that purpose. Hence

$$\kappa \stackrel{\text{def}}{=} \left. \frac{\partial (\eta_1, \eta_2, \eta_3)}{\partial (r'_1, r'_2, r'_3)} \right|_{\boldsymbol{\eta}=\mathbf{0}} = 1 - \frac{\mathbf{v}(t')}{c} \cdot \mathbf{e}_{r^*}. \quad (3.36)$$

Once we know what $\mathbf{f}^{-1}(\mathbf{0})$ and $\left. \frac{\partial (\eta_1, \eta_2, \eta_3)}{\partial (r'_1, r'_2, r'_3)} \right|_{\boldsymbol{\eta}=\mathbf{0}}$ are, we may finally write for the potentials

$$\begin{aligned}
\phi(\mathbf{r}, t) &= \frac{q}{4\pi\epsilon_0} \frac{1}{\left| \mathbf{r} - \boldsymbol{\zeta}(t') \right| \left[1 - \mathbf{v}(t') \frac{\mathbf{e}_{r^*}}{c} \right]} = \frac{q}{4\pi\epsilon_0} \frac{1}{r^* - \frac{\mathbf{v}(t') \cdot \mathbf{r}^*}{c}} \\
&= \frac{q}{4\pi\epsilon_0 r^*} \frac{1}{1 - \mathbf{v}(t') \frac{\mathbf{e}_{r^*}}{c}} = \frac{q}{4\pi\epsilon_0 r^*} \frac{1}{\kappa}, \quad (3.37)
\end{aligned}$$

$$\begin{aligned}
\mathbf{A}(\mathbf{r}, t) &= \frac{q}{4\pi\epsilon_0 c^2} \frac{\mathbf{v}(t')}{\left| \mathbf{r} - \boldsymbol{\zeta}(t') \right| \left[1 - \mathbf{v}(t') \frac{\mathbf{e}_{r^*}}{c} \right]} = \frac{q}{4\pi\epsilon_0 c^2} \frac{\mathbf{v}(t')}{r^* - \frac{\mathbf{v}(t') \cdot \mathbf{r}^*}{c}} \\
&= \frac{q}{4\pi\epsilon_0 c^2 r^*} \frac{\mathbf{v}(t')}{1 - \mathbf{v}(t') \frac{\mathbf{e}_{r^*}}{c}} = \frac{q}{4\pi\epsilon_0 c^2 r^*} \frac{\mathbf{v}(t')}{\kappa}. \quad (3.38)
\end{aligned}$$

The potentials (3.37) and (3.38) are called Liénard-Wiechert potentials and describe scalar $\phi(\mathbf{r}, t)$ and vector $\mathbf{A}(\mathbf{r}, t)$ potentials of a point charge moving with arbitrary velocity $\mathbf{v}$. $\mathbf{r}^* = \mathbf{r} - \boldsymbol{\zeta}(t')$ is the vector from the charge position $\boldsymbol{\zeta}(t')$ at the retarded time $t' = t - r^*/c$ to the point of observation $\mathbf{r}$.

4 Fields of a point charge moving with arbitrary velocity

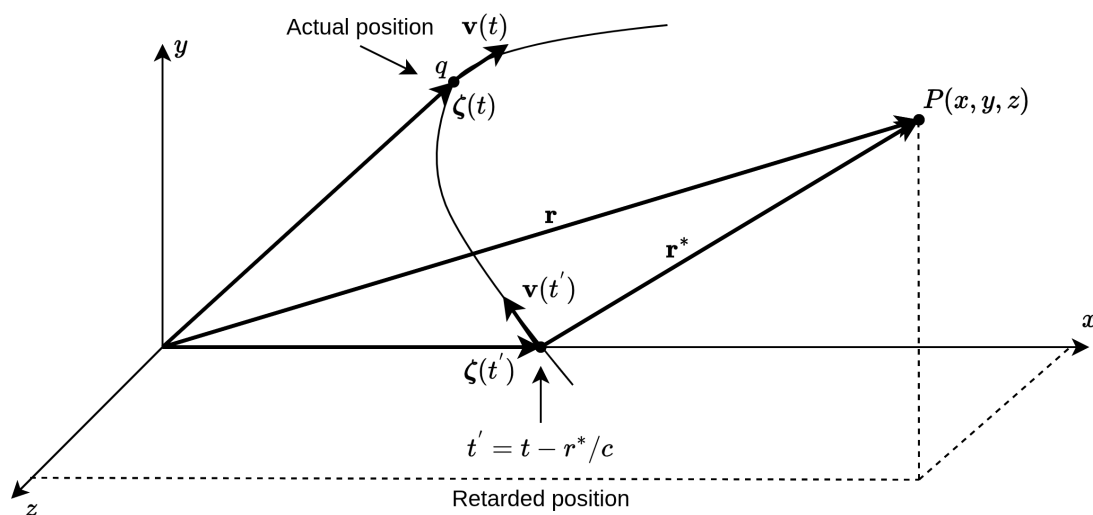


Figure 4.1: Point charge moving with arbitrary velocity.

4.1 Electric and magnetic fields

In the previous chapter we have derived expressions (3.37) and (3.38) which are called Liénard–Wiechert potentials and describe scalar $\phi(\mathbf{r}, t)$ and vector $\mathbf{A}(\mathbf{r}, t)$ potentials of a point charge moving with arbitrary velocity $\mathbf{v}$. Having got that potentials we may now evaluate electric and magnetic fields of a point charge moving with arbitrary velocity. To do so we will use already very well known expressions:

$$\mathbf{E} = -\nabla\phi - \frac{\partial\mathbf{A}}{\partial t} \quad (4.1)$$

$$\mathbf{B} = \nabla \times \mathbf{A} \quad (4.2)$$

Let us first evaluate the gradient of the scalar potential $\nabla\phi$. Recall that the Liénard–Wiechert scalar potential is given by

$$\phi(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0} \frac{1}{\kappa r^*}, \quad (4.3)$$

where $r^* = |\mathbf{r} - \boldsymbol{\zeta}(t')|$ is the distance from the field point to the retarded position of the charge, and $\kappa = 1 - \frac{\mathbf{v}(t')}{c} \cdot \mathbf{e}_{r^*}$.

The gradient with respect to $\mathbf{r}$ acts both on r^* and on κ . Thus

$$\begin{aligned} \nabla \phi &= \frac{q}{4\pi\epsilon_0} \nabla \left(\frac{1}{\kappa r^*} \right) = \frac{q}{4\pi\epsilon_0} \left[-\frac{1}{(\kappa r^*)^2} \nabla(\kappa r^*) \right] = \\ &- \frac{q}{4\pi\epsilon_0} \frac{1}{(\kappa r^*)^2} \nabla \left(r^* - \frac{\mathbf{v}(t') \cdot \mathbf{r}^*}{c} \right) = -\frac{q}{4\pi\epsilon_0} \frac{1}{(\kappa r^*)^2} \left[\nabla r^* - \frac{1}{c} \nabla (\mathbf{v}(t') \cdot \mathbf{r}^*) \right] \end{aligned} \quad (4.4)$$

We see that to evaluate $\nabla \phi$ we need to evaluate ∇r^* and $\nabla (\mathbf{v}(t') \cdot \mathbf{r}^*)$. Let's start this rather lengthy process with the evaluation of ∇r^* .

$$\nabla r^* = \left(\frac{\partial r^*}{\partial r_1}, \frac{\partial r^*}{\partial r_2}, \frac{\partial r^*}{\partial r_3} \right) = \frac{\partial r^*}{\partial r_i} \mathbf{e}_i \quad (4.5)$$

where

$$\frac{\partial r^*}{\partial r_i} = \frac{1}{2r^*} \frac{\partial r^{*2}}{\partial r_i} = \frac{1}{2r^*} \frac{\partial (\mathbf{r}^* \cdot \mathbf{r}^*)}{\partial r_i} = \frac{1}{2r^*} \frac{2\mathbf{r}^* \partial \mathbf{r}^*}{\partial r_i} = \frac{\mathbf{r}^* \partial \mathbf{r}^*}{r^* \partial r_i} = \mathbf{e}_{r^*} \cdot \frac{\partial \mathbf{r}^*}{\partial r_i} \quad (4.6)$$

and $\mathbf{e}_i$ are the orthogonal unit vectors of the Cartesian coordinate system. Now, we may recall that $\mathbf{r}^* = \mathbf{r} - \boldsymbol{\zeta}(t')$. Hence

$$\frac{\partial \mathbf{r}^*}{\partial r_i} = \frac{\partial (\mathbf{r} - \boldsymbol{\zeta}(t'))}{\partial r_i} = \frac{\partial \mathbf{r}}{\partial r_i} - \frac{\partial \boldsymbol{\zeta}(t')}{\partial r_i} = \mathbf{e}_i - \frac{d\boldsymbol{\zeta}(t')}{dt'} \frac{\partial t'}{\partial r_i} = \mathbf{e}_i - \mathbf{v}(t') \frac{\partial t'}{\partial r_i} \quad (4.7)$$

Since retarded time $t' = t - r^*/c$,

$$\nabla t' = -\frac{1}{c} \nabla r^* \quad (4.8)$$

or, which is equivalent

$$\frac{\partial t'}{\partial r_i} \mathbf{e}_i = -\frac{1}{c} \frac{\partial r^*}{\partial r_i} \mathbf{e}_i. \quad (4.9)$$

With all those calculations, we may finally write.

$$\frac{\partial r^*}{\partial r_i} = \mathbf{e}_{r^*} \cdot \left(\mathbf{e}_i - \mathbf{v}(t') \frac{\partial t'}{\partial r_i} \right) = \mathbf{e}_{r^*} \cdot \mathbf{e}_i + \frac{\mathbf{e}_{r^*} \cdot \mathbf{v}(t')}{c} \frac{\partial r^*}{\partial r_i} \quad (4.10)$$

Thus

$$\frac{\partial r^*}{\partial r_i} = \frac{\mathbf{e}_{r^*} \cdot \mathbf{e}_i}{1 - \frac{\mathbf{v}(t')}{c} \cdot \mathbf{e}_{r^*}} = \frac{\mathbf{e}_{r^*} \cdot \mathbf{e}_i}{\kappa} \quad (4.11)$$

and finally

$$\nabla r^* = \frac{\mathbf{e}_{r^*}}{\kappa} = \frac{\mathbf{r}^*}{\kappa r^*}. \quad (4.12)$$

Okay, we have now the expression for ∇r^* . What is left, is to find the expression for $\nabla (\mathbf{v}(t') \cdot \mathbf{r}^*)$. Let us stick with that in the following lines (or even pages as the process can be a bit painful).

We see that to calculate $\nabla (\mathbf{v}(t') \cdot \mathbf{r}^*)$ we shall calculate gradient of a dot product of two vector fields $\mathbf{v}(t')$ and $\mathbf{r}^*$. And the vector calculus identity for that is as follows:

$$\nabla (\mathbf{a} \cdot \mathbf{b}) = (\mathbf{a} \cdot \nabla) \mathbf{b} + (\mathbf{b} \cdot \nabla) \mathbf{a} + \mathbf{a} \times (\nabla \times \mathbf{b}) + \mathbf{b} \times (\nabla \times \mathbf{a}). \quad (4.13)$$

Substituting $\mathbf{v}(t')$ and $\mathbf{r}^*$ for $\mathbf{a}$ and $\mathbf{b}$, we get

$$\begin{aligned} \nabla (\mathbf{v}(t') \cdot \mathbf{r}^*) &= (\mathbf{v}(t') \cdot \nabla) \mathbf{r}^* + (\mathbf{r}^* \cdot \nabla) \mathbf{v}(t') + \\ &\quad \mathbf{v}(t') \times (\nabla \times \mathbf{r}^*) + \mathbf{r}^* \times (\nabla \times \mathbf{v}(t')). \end{aligned} \quad (4.14)$$

Let us evaluate each term in the expression (4.14) separately:

1. First term: $(\mathbf{v}(t') \cdot \nabla) \mathbf{r}^*$

Since $\mathbf{r}^* = \mathbf{r} - \boldsymbol{\zeta}(t')$, and $\mathbf{v}(t')$ is evaluated at the retarded time (which depends on $\mathbf{r}$), we have

$$(\mathbf{v}(t') \cdot \nabla) \mathbf{r}^* = (\mathbf{v}(t') \cdot \nabla) \mathbf{r} - (\mathbf{v}(t') \cdot \nabla) \boldsymbol{\zeta}(t') \quad (4.15)$$

The first term is simply $\mathbf{v}(t')$, since $\nabla \mathbf{r} = \mathbf{I}$ (identity), and the second term is more subtle, but since $\boldsymbol{\zeta}(t')$ depends on t' , and t' depends on $\mathbf{r}$, we get

$$\begin{aligned} (\mathbf{v}(t') \cdot \nabla) \boldsymbol{\zeta}(t') &= v_i(t') \frac{\partial \boldsymbol{\zeta}(t')}{\partial r_i} \mathbf{e}_i = v_i(t') \frac{d\boldsymbol{\zeta}(t')}{dt'} \frac{\partial t'}{\partial r_i} \mathbf{e}_i = \\ &= \frac{d\boldsymbol{\zeta}(t')}{dt'} \left(v_i(t') \frac{\partial t'}{\partial r_i} \mathbf{e}_i \right) = \mathbf{v}(t') (\mathbf{v}(t') \cdot \nabla t') \end{aligned} \quad (4.16)$$

Therefore,

$$\left(\mathbf{v}(t') \cdot \nabla\right) \mathbf{r}^* = \mathbf{v}(t') - \mathbf{v}(t') \left(\mathbf{v}(t') \cdot \nabla t'\right) \quad (4.17)$$

2. **Second term:** $(\mathbf{r}^* \cdot \nabla) \mathbf{v}(t')$

Since $\mathbf{v}(t')$ depends on t' , and t' depends on $\mathbf{r}$,

$$\begin{aligned} (\mathbf{r}^* \cdot \nabla) \mathbf{v}(t') &= r_i^* \frac{\partial \mathbf{v}(t')}{\partial r_i} \mathbf{e}_i = r_i^* \frac{d\mathbf{v}(t')}{dt'} \frac{\partial t'}{\partial r_i} \mathbf{e}_i = \\ &= \frac{d\mathbf{v}_i(t')}{dt'} \left(r_i^* \frac{\partial t'}{\partial r_i} \mathbf{e}_i \right) = \mathbf{a}(t') \left(\mathbf{r}^* \cdot \nabla t' \right) \end{aligned} \quad (4.18)$$

where $\mathbf{a}(t') = \frac{d\mathbf{v}(t')}{dt'}$ is the acceleration of the charge at the retarded time t' .

3. **Third term:** $\mathbf{v}(t') \times (\nabla \times \mathbf{r}^*)$

Since $\mathbf{r}^* = \mathbf{r} - \boldsymbol{\zeta}(t')$, we have

$$\mathbf{v}(t') \times (\nabla \times \mathbf{r}^*) = \mathbf{v}(t') \times \left(\nabla \times \mathbf{r} - \nabla \times \boldsymbol{\zeta}(t') \right). \quad (4.19)$$

As $\mathbf{r}$ is a position vector, $\nabla \times \mathbf{r} = 0$, and $\nabla \times \mathbf{r}$ term vanishes, whereas $\nabla \times \boldsymbol{\zeta}(t')$ is evaluated as follows:

$$\begin{aligned} \nabla \times \boldsymbol{\zeta}(t') &= \\ \left(\frac{\partial \zeta_3(t')}{\partial r_2} - \frac{\partial \zeta_2(t')}{\partial r_3} \right) \mathbf{e}_1 + \left(\frac{\partial \zeta_1(t')}{\partial r_3} - \frac{\partial \zeta_3(t')}{\partial r_1} \right) \mathbf{e}_2 + \left(\frac{\partial \zeta_2(t')}{\partial r_1} - \frac{\partial \zeta_1(t')}{\partial r_2} \right) \mathbf{e}_3 &= \\ \left(\frac{d\zeta_3(t')}{dt'} \frac{\partial t'}{\partial r_2} - \frac{d\zeta_2(t')}{dt'} \frac{\partial t'}{\partial r_3} \right) \mathbf{e}_1 + & \\ \left(\frac{d\zeta_1(t')}{dt'} \frac{\partial t'}{\partial r_3} - \frac{d\zeta_3(t')}{dt'} \frac{\partial t'}{\partial r_1} \right) \mathbf{e}_2 + & \\ \left(\frac{d\zeta_2(t')}{dt'} \frac{\partial t'}{\partial r_1} - \frac{d\zeta_1(t')}{dt'} \frac{\partial t'}{\partial r_2} \right) \mathbf{e}_3 &= \\ -\mathbf{v}(t') \times \nabla t'. & \end{aligned}$$

Therefore,

$$\mathbf{v}(t') \times (\nabla \times \mathbf{r}^*) = \mathbf{v}(t') \times \left(\mathbf{v}(t') \times \nabla t' \right) \quad (4.20)$$

4. **Fourth term:** $\mathbf{r}^* \times (\nabla \times \mathbf{v}(t'))$

Using the same procedure as in third term for $\nabla \times \boldsymbol{\zeta}(t')$, we write

$$\nabla \times \mathbf{v}(t') = -\mathbf{a}(t') \times \nabla t' \quad (4.21)$$

and the whole term is

$$\mathbf{r}^* \times (\nabla \times \mathbf{v}(t')) = -\mathbf{r}^* \times (\mathbf{a}(t') \times \nabla t') \quad (4.22)$$

Adding all those 4 terms together, equation (4.14) becomes

$$\begin{aligned} \nabla (\mathbf{v}(t') \cdot \mathbf{r}^*) &= \mathbf{v}(t') - \mathbf{v}(t') (\mathbf{v}(t') \cdot \nabla t') + \mathbf{a}(t') (\mathbf{r}^* \cdot \nabla t') \\ &\quad + \mathbf{v}(t') \times (\mathbf{v}(t') \times \nabla t') - \mathbf{r}^* \times (\mathbf{a}(t') \times \nabla t') \end{aligned} \quad (4.23)$$

To get rid of cross products in (4.23) we may use yet another vector calculus identity which is usually called "bac-cab". Here it is.

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - \mathbf{c}(\mathbf{a} \cdot \mathbf{b}) \quad (4.24)$$

Applying this identity to the cross product terms in (4.23), we get:

$$\begin{aligned} \mathbf{v}(t') \times (\mathbf{v}(t') \times \nabla t') &= \mathbf{v}(t') (\mathbf{v}(t') \cdot \nabla t') - \nabla t' (\mathbf{v}(t') \cdot \mathbf{v}(t')) \\ \mathbf{r}^* \times (\mathbf{a}(t') \times \nabla t') &= \mathbf{a}(t') (\mathbf{r}^* \cdot \nabla t') - \nabla t' (\mathbf{r}^* \cdot \mathbf{a}(t')) \end{aligned}$$

Substituting these into (4.23), we obtain:

$$\begin{aligned} \nabla (\mathbf{v}(t') \cdot \mathbf{r}^*) &= \mathbf{v}(t') - \mathbf{v}(t') (\mathbf{v}(t') \cdot \nabla t') + \mathbf{a}(t') (\mathbf{r}^* \cdot \nabla t') \\ &\quad + \mathbf{v}(t') (\mathbf{v}(t') \cdot \nabla t') - \nabla t' (\mathbf{v}(t') \cdot \mathbf{v}(t')) - \mathbf{a}(t') (\mathbf{r}^* \cdot \nabla t') + \nabla t' (\mathbf{r}^* \cdot \mathbf{a}(t')) \end{aligned}$$

Notice that the terms $\mathbf{v}(t') (\mathbf{v}(t') \cdot \nabla t')$ and $\mathbf{a}(t') (\mathbf{r}^* \cdot \nabla t')$ cancel, so the expression (4.23) simplifies to:

$$\nabla (\mathbf{v}(t') \cdot \mathbf{r}^*) = \mathbf{v}(t') + (\mathbf{r}^* \cdot \mathbf{a}(t') - v^2(t')) \nabla t' \quad (4.25)$$

This form is more convenient for further simplification and for substituting back into the expression for $\nabla \phi$. Let's do so then and substitute (4.25) into (4.4):

$$\nabla\phi = -\frac{q}{4\pi\epsilon_0} \frac{1}{(\kappa r^*)^2} \left[\nabla r^* - \frac{1}{c} \left(\mathbf{v}(t') + \left(\mathbf{r}^* \cdot \mathbf{a}(t') - v^2(t') \right) \nabla t' \right) \right] \quad (4.26)$$

Substituting $-c\nabla t'$ for ∇r^* (see equation (4.8)) and rearranging terms in (4.26) we get

$$\nabla\phi = \frac{q}{4\pi\epsilon_0} \frac{1}{(\kappa r^*)^2} \left[\frac{\mathbf{v}(t')}{c} + \frac{1}{c} \left(c^2 + \mathbf{r}^* \cdot \mathbf{a}(t') - v^2(t') \right) \nabla t' \right]. \quad (4.27)$$

And finally, substituting $-\frac{1}{c} \frac{\mathbf{r}^*}{\kappa r^*}$ for $\nabla t'$ (see (4.12) and (4.8)) we get the expression for the gradient $\nabla\phi$ of the Liénard–Wiechert scalar potential for a point charge moving with arbitrary velocity $\mathbf{v}(t')$

$$\nabla\phi = \frac{q}{4\pi\epsilon_0} \frac{1}{(\kappa r^*)^3} \left[\frac{\mathbf{v}(t')\kappa r^*}{c} - \frac{1}{c^2} \left(c^2 + \mathbf{r}^* \cdot \mathbf{a}(t') - v^2(t') \right) \mathbf{r}^* \right]. \quad (4.28)$$

To calculate $\mathbf{E} = -\nabla\phi - \frac{\partial\mathbf{A}}{\partial t}$ we are missing yet expression for $\frac{\partial\mathbf{A}}{\partial t}$. Let's solve that missing part now.

We will start this process by recalling Liénard–Wiechert vector potential

$$\mathbf{A}(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0 c^2} \frac{\mathbf{v}(t')}{\kappa r^*}, \quad (4.29)$$

where all quantities are evaluated at the retarded time t' .

To compute $\frac{\partial\mathbf{A}}{\partial t}$, note that both $\mathbf{v}(t')$, κ , and r^* depend on t via t' :

$$\frac{\partial\mathbf{A}}{\partial t} = \frac{q}{4\pi\epsilon_0 c^2} \frac{\partial}{\partial t} \left(\frac{\mathbf{v}(t')}{\kappa r^*} \right) \quad (4.30)$$

Using the product and chain rules:

$$\frac{\partial}{\partial t} \left(\frac{\mathbf{v}(t')}{\kappa r^*} \right) = \frac{1}{\kappa r^*} \frac{d\mathbf{v}(t')}{dt'} \frac{\partial t'}{\partial t} - \frac{\mathbf{v}(t')}{(\kappa r^*)^2} \frac{\partial}{\partial t} (\kappa r^*) \quad (4.31)$$

Now, $\frac{d\mathbf{v}(t')}{dt'} = \mathbf{a}(t')$, and since $t' = t - \frac{r^*}{c}$, $\frac{\partial t'}{\partial t} = 1 - \frac{1}{c} \frac{\partial r^*}{\partial t}$.

For $\frac{\partial r^*}{\partial t}$ let us remind that $\boldsymbol{\zeta}(t') = \mathbf{r} - \mathbf{r}^*$. Then

$$\frac{\partial \boldsymbol{\zeta}(t')}{\partial t} = -\frac{\partial \mathbf{r}^*}{\partial t} \quad (4.32)$$

which further yields

$$\frac{d\boldsymbol{\zeta}(t')}{dt'} \frac{\partial t'}{\partial t} = -\frac{\partial(r^* \mathbf{e}_{r^*})}{\partial t} = -\left(\frac{\partial r^*}{\partial t} \mathbf{e}_{r^*} + r^* \frac{\partial \mathbf{e}_{r^*}}{\partial t} \right). \quad (4.33)$$

Substituting $\mathbf{v}(t')$ for $\frac{d\boldsymbol{\zeta}(t')}{dt'}$ and $1 - \frac{1}{c} \frac{\partial r^*}{\partial t}$ for $\frac{\partial t'}{\partial t}$ we write

$$\mathbf{v}(t') = -\frac{\frac{\partial r^*}{\partial t} \mathbf{e}_{r^*} + r^* \frac{\partial \mathbf{e}_{r^*}}{\partial t}}{1 - \frac{1}{c} \frac{\partial r^*}{\partial t}}. \quad (4.34)$$

And now, substituting $\mathbf{v}(t')$ into expression defining κ ($\kappa = 1 - \frac{\mathbf{v}(t') \cdot \mathbf{e}_{r^*}}{c}$), we write for κ

$$\kappa = \frac{1}{1 - \frac{1}{c} \frac{\partial r^*}{\partial t}}. \quad (4.35)$$

Please note that

$$\frac{\partial \mathbf{e}_{r^*}}{\partial t} \cdot \mathbf{e}_{r^*} = \frac{1}{2} \frac{\partial (\mathbf{e}_{r^*} \cdot \mathbf{e}_{r^*})}{\partial t} = \frac{1}{2} \frac{\partial |\mathbf{e}_{r^*}|^2}{\partial t} = 0. \quad (4.36)$$

After straightforward transformation of (4.35) we get:

$$\frac{\partial r^*}{\partial t} = \frac{\kappa - 1}{\kappa} c. \quad (4.37)$$

So,

$$\frac{\partial t'}{\partial t} = 1 - \frac{1}{c} \frac{\partial r^*}{\partial t} = \frac{1}{\kappa} \quad (4.38)$$

Now, for $\frac{\partial}{\partial t}(\kappa r^*)$:

$$\frac{\partial}{\partial t}(\kappa r^*) = \frac{\partial r^*}{\partial t} - \frac{1}{c} \frac{\partial}{\partial t}(\mathbf{v}(t') \cdot \mathbf{r}^*). \quad (4.39)$$

$$\frac{\partial}{\partial t}(\mathbf{v}(t') \cdot \mathbf{r}^*) = \frac{d\mathbf{v}(t')}{dt'} \frac{\partial t'}{\partial t} \cdot \mathbf{r}^* + \mathbf{v}(t') \cdot \frac{\partial \mathbf{r}^*}{\partial t} \quad (4.40)$$

$$= \mathbf{a}(t') \cdot \mathbf{r}^* \frac{\partial t'}{\partial t} + \mathbf{v}(t') \cdot \left(-\mathbf{v}(t') \frac{\partial t'}{\partial t} \right) \quad (4.41)$$

$$= \left(\mathbf{r}^* \cdot \mathbf{a}(t') - v^2(t') \right) \frac{\partial t'}{\partial t}. \quad (4.42)$$

Substituting $\frac{1}{\kappa}$ for $\frac{\partial t'}{\partial t}$ and $\frac{\kappa-1}{\kappa}c$ for $\frac{\partial r^*}{\partial t}$ we finally get for $\frac{\partial}{\partial t}(\kappa r^*)$:

$$\frac{\partial}{\partial t}(\kappa r^*) = \frac{1}{\kappa} \left[(\kappa-1)c - \frac{1}{c} \left(\mathbf{r}^* \cdot \mathbf{a}(t') - v^2(t') \right) \right]. \quad (4.43)$$

And now, substituting $\frac{\partial}{\partial t}(\kappa r^*)$ into (4.31) we may write

$$\begin{aligned} \frac{\partial}{\partial t} \left(\frac{\mathbf{v}(t')}{\kappa r^*} \right) &= \frac{1}{\kappa r^*} \mathbf{a}(t') \frac{1}{\kappa} - \frac{\mathbf{v}(t')}{(\kappa r^*)^2} \frac{1}{\kappa} \left[(\kappa-1)c - \frac{1}{c} \left(\mathbf{r}^* \cdot \mathbf{a}(t') - v^2(t') \right) \right] = \\ &= \frac{1}{(\kappa r^*)^3} \left[\kappa r^{*2} \mathbf{a}(t') - \mathbf{v}(t') r^* \kappa c + \mathbf{v}(t') r^* c + \frac{\mathbf{v}(t') r^*}{c} \left(\mathbf{r}^* \cdot \mathbf{a}(t') - v^2(t') \right) \right] = \\ &= \frac{1}{(\kappa r^*)^3} \left[\kappa r^* \left(r^* \mathbf{a}(t') - \mathbf{v}(t') c \right) + \frac{r^*}{c} \left(c^2 + \mathbf{r}^* \cdot \mathbf{a}(t') - v^2(t') \right) \mathbf{v}(t') \right]. \end{aligned}$$

Thus

$$\frac{\partial \mathbf{A}}{\partial t} = \frac{q}{4\pi\epsilon_0} \frac{1}{(\kappa r^*)^3} \left[\frac{\kappa r^*}{c^2} \left(r^* \mathbf{a}(t') - \mathbf{v}(t') c \right) + \frac{r^*}{c^3} \left(c^2 + \mathbf{r}^* \cdot \mathbf{a}(t') - v^2(t') \right) \mathbf{v}(t') \right]. \quad (4.44)$$

Adding expressions for $\nabla\phi$ (4.28) and $\frac{\partial \mathbf{A}}{\partial t}$ (4.44) we finally may write equation for the electric field of a point charge moving with arbitrary velocity $\mathbf{v}$:

$$\begin{aligned} \mathbf{E}(\mathbf{r}, t) &= -\nabla\phi - \frac{\partial \mathbf{A}}{\partial t} = -\frac{q}{4\pi\epsilon_0} \frac{1}{(\kappa r^*)^3} \left[\frac{\mathbf{v}(t') \kappa r^*}{c} - \right. \\ &\quad \left. \frac{1}{c^2} \left(c^2 + \mathbf{r}^* \cdot \mathbf{a}(t') - v^2(t') \right) \mathbf{r}^* + \frac{\kappa r^*}{c^2} \left(r^* \mathbf{a}(t') - \mathbf{v}(t') c \right) + \right. \\ &\quad \left. \frac{r^*}{c^3} \left(c^2 + \mathbf{r}^* \cdot \mathbf{a}(t') - v^2(t') \right) \mathbf{v}(t') \right] = \\ &= -\frac{q}{4\pi\epsilon_0} \frac{1}{(\kappa r^*)^3} \left[\left(c^2 + \mathbf{r}^* \cdot \mathbf{a}(t') - v^2(t') \right) \left(\frac{r^* \mathbf{v}(t')}{c^3} - \frac{\mathbf{r}^*}{c^2} \right) + \frac{\kappa r^{*2}}{c^2} \mathbf{a}(t') \right] = \\ &= \frac{q}{4\pi\epsilon_0} \frac{r^*}{(\kappa r^*)^3} \left[\left(c^2 + \mathbf{r}^* \cdot \mathbf{a}(t') - v^2(t') \right) \left(\frac{c \mathbf{e}_{r^*}}{c^3} - \frac{\mathbf{v}(t')}{c^3} \right) - \frac{c \kappa r^*}{c^3} \mathbf{a}(t') \right] \quad (4.45) \end{aligned}$$

At this point we will introduce a helper vector

$$\mathbf{u}(t') \stackrel{\text{def}}{=} c \mathbf{e}_{r^*} - \mathbf{v}(t'). \quad (4.46)$$

With that helper vector, $c\kappa r^* = \mathbf{r}^* \cdot \mathbf{u}$, and our equation for the electric field gains the following shape

$$\mathbf{E}(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0} \frac{r^*}{(\mathbf{r}^* \cdot \mathbf{u}(t'))^3} \left[\left(c^2 - v^2(t') \right) \mathbf{u}(t') + \left(\mathbf{r}^* \cdot \mathbf{a}(t') \right) \mathbf{u}(t') - \left(\mathbf{r}^* \cdot \mathbf{u}(t') \right) \mathbf{a}(t') \right]. \quad (4.47)$$

We may still rewrite the above equation in a neater form using already introduced "bac-cab" identity (see (4.24)). Doing so, we get

$$\mathbf{E}(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0} \frac{r^*}{(\mathbf{r}^* \cdot \mathbf{u}(t'))^3} \left[\left(c^2 - v^2(t') \right) \mathbf{u}(t') + \mathbf{r}^* \times \left(\mathbf{u}(t') \times \mathbf{a}(t') \right) \right]. \quad (4.48)$$

Wow, several pages and we just calculated $\mathbf{E}$. We are still missing the second part of these puzzles, which is $\mathbf{B}$. But let us stop for a moment here before we go any further with derivations of $\mathbf{B}$.

The equation (4.48) shows that the electric field consists of two terms: the one involving $\left(c^2 - v^2(t') \right) \mathbf{u}(t')$ that falls off as $1/r^{*2}$ (the "velocity" or "Coulomb-like" field), and one (involving $\mathbf{r}^* \times \left(\mathbf{u}(t') \times \mathbf{a}(t') \right)$) that falls off as $1/r^*$ (the "acceleration" or "radiation" field). The $1/r^*$ term dominates at large distances and is responsible for phenomena which we call electromagnetic radiation. And I believe this is the first time the word "radiation" appeared in this paper although it might seem that it shall be appearing much more often as the paper is titled "Radiation by moving charges". But don't worry, soon we will be using it very often. Especially when discussing Poynting vector.

Let us come back now to the derivations of the magnetic field $\mathbf{B}$ produced by a point charge moving with arbitrary velocity. We will start that process by recalling equation (4.2)

$$\mathbf{B} = \nabla \times \mathbf{A}.$$

Because

$$\mathbf{A}(\mathbf{r}, t) = \frac{1}{c^2} \mathbf{v}(t') \phi(\mathbf{r}, t)$$

we write

$$\mathbf{B}(\mathbf{r}, t) = \frac{1}{c^2} \nabla \times \left(\mathbf{v}(t') \phi(\mathbf{r}, t) \right) = \frac{1}{c^2} \left[\phi(\mathbf{r}, t) \left(\nabla \times \mathbf{v}(t') \right) + \mathbf{v}(t') \times \left(\nabla \phi(\mathbf{r}, t) \right) \right].$$

But hang on, we have already calculated $\nabla \times \mathbf{v}(t')$ (see (4.21), (4.8), (4.12)) and $\nabla\phi(\mathbf{r}, t)$ (see (4.28)) while evaluating electric field. Let's use those results here and continue with derivations of $\mathbf{B}$

$$\mathbf{B}(\mathbf{r}, t) = \frac{1}{c} \frac{q}{4\pi\epsilon_0} \frac{1}{(c\kappa r^*)^3} \left\{ c\kappa r^* \left(\mathbf{a}(t') \times \mathbf{r}^* \right) + \mathbf{v}(t') \times \left[\mathbf{v}(t') c\kappa r^* - \left(c^2 + \mathbf{r}^* \cdot \mathbf{a}(t') - v^2(t') \right) \mathbf{r}^* \right] \right\}.$$

We will one more time use our helper vector $\mathbf{u}$, and yet again noting that $c\kappa r^* = \mathbf{r}^* \cdot \mathbf{u}$, the expression for $\mathbf{B}$ gains the following form

$$\mathbf{B}(\mathbf{r}, t) = \frac{1}{c} \frac{q}{4\pi\epsilon_0} \frac{1}{(\mathbf{r}^* \cdot \mathbf{u}(t'))^3} \left\{ (\mathbf{r}^* \cdot \mathbf{u}) \left(\mathbf{a}(t') \times \mathbf{r}^* \right) + \mathbf{v}(t') \times \left[\left(c^2 + \mathbf{r}^* \cdot \mathbf{a}(t') - v^2(t') \right) \mathbf{r}^* \right] \right\}$$

which after rearrangements can be written as

$$\mathbf{B}(\mathbf{r}, t) = -\frac{1}{c} \frac{q}{4\pi\epsilon_0} \frac{1}{(\mathbf{r}^* \cdot \mathbf{u}(t'))^3} \left\{ \mathbf{r}^* \times \left[\left(c^2 - v^2(t') \right) \mathbf{v}(t') + \left(\mathbf{r}^* \cdot \mathbf{a}(t') \right) \mathbf{v}(t') + \left(\mathbf{r}^* \cdot \mathbf{u}(t') \right) \mathbf{a}(t') \right] \right\}.$$

If we now note that

$$\mathbf{r}^* \times \mathbf{v}(t') = \mathbf{r}^* \times \left(c\mathbf{e}_{r^*} - \mathbf{u}(t') \right) = -\mathbf{r}^* \times \mathbf{u}(t') \quad (4.49)$$

we may further write

$$\mathbf{B}(\mathbf{r}, t) = \frac{1}{c} \frac{q}{4\pi\epsilon_0} \frac{1}{(\mathbf{r}^* \cdot \mathbf{u}(t'))^3} \left\{ \mathbf{r}^* \times \left[\left(c^2 - v^2(t') \right) \mathbf{u}(t') + \left(\mathbf{r}^* \cdot \mathbf{a}(t') \right) \mathbf{u}(t') - \left(\mathbf{r}^* \cdot \mathbf{u}(t') \right) \mathbf{a}(t') \right] \right\}$$

which if we look carefully into (4.47) can be written as

$$\mathbf{B}(\mathbf{r}, t) = \frac{1}{c} \mathbf{e}_{r^*} \times \mathbf{E}(\mathbf{r}, t). \quad (4.50)$$

Summarizing all that we have learned, the equations for the electric $\mathbf{E}$ and the magnetic $\mathbf{B}$ fields of a point charge moving with arbitrary velocity $\mathbf{v}$ can be expressed using the following formulas:

$$\mathbf{E}(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0} \frac{r^*}{(\mathbf{r}^* \cdot \mathbf{u}(t'))^3} \left[(c^2 - v^2(t')) \mathbf{u}(t') + \mathbf{r}^* \times (\mathbf{u}(t') \times \mathbf{a}(t')) \right] \quad (4.51)$$

$$\mathbf{B}(\mathbf{r}, t) = \frac{1}{c} \mathbf{e}_{r^*} \times \mathbf{E}(\mathbf{r}, t) \quad (4.52)$$

4.2 The Poynting vector

Just by definition the Poynting vector is

$$\begin{aligned} \mathbf{S}(\mathbf{r}, t) &= \mathbf{E}(\mathbf{r}, t) \times \mathbf{H}(\mathbf{r}, t) = c^2 \epsilon_0 \mathbf{E}(\mathbf{r}, t) \times \mathbf{B}(\mathbf{r}, t) \\ &= c \epsilon_0 \mathbf{E}(\mathbf{r}, t) \times (\mathbf{e}_{r^*} \times \mathbf{E}(\mathbf{r}, t)). \end{aligned} \quad (4.53)$$

If we now recall our considerations on the subject of "velocity" and "acceleration" terms in the formula for the electric field (4.48), looking at equation (4.53), we may further conclude that Poynting vector $\mathbf{S}$ has three terms. The one which falls off as $1/r^{*4}$, another one which falls off as $1/r^{*3}$ and the last one which falls off as $1/r^{*2}$. Naturally, the $1/r^{*4}$ term dominates at very small distances from the retarded position of the charge, but the same it is vanishingly small far away from the charge. On the other hand the $1/r^{*2}$ term dominates at far distances from the charge, whereas it can be neglected when discussing very small charge-observer distances. To make things easier and stick only on important terms when discussing different aspects of the EM propagation, we call the $1/r^{*4}$ term a "near field" term and use it when describing EM field very close to the charge, whereas the $1/r^{*2}$ is called a "far field" or "radiation" term and is used when discussing EM radiation far away from the moving charge which is the source of radiation. If we look into "near field" and "far field" terms one more time, we will also notice, that only "far field" term depends on the acceleration of the charge. That is why we say, that only accelerating charged particles produce radiation. Static charges or charges moving with constant velocity do not radiate.

Using identity (4.24) we may rewrite equation (4.53) as

$$\mathbf{S}(\mathbf{r}, t) = c \epsilon_0 \mathbf{E}^2(\mathbf{r}, t) \mathbf{e}_{r^*} - c \epsilon_0 \mathbf{E}(\mathbf{r}, t) (\mathbf{E}(\mathbf{r}, t) \cdot \mathbf{e}_{r^*}) \quad (4.54)$$

and noting that radiation term has $\mathbf{E}$ which is all the time perpendicular to $\mathbf{e}_{r^*}$, $\mathbf{S}$ reduces to the following expression

$$\mathbf{S}(\mathbf{r}, t) = c \epsilon_0 \mathbf{E}^2(\mathbf{r}, t) \mathbf{e}_{r^*}. \quad (4.55)$$

Expression (4.55) can be written as

$$\mathbf{S}(\mathbf{r}, t) = c \frac{q^2}{16\pi^2\epsilon_0} \frac{r^{*2}}{\left(\mathbf{r}^* \cdot \mathbf{u}(t')\right)^6} \left[\mathbf{r}^* \times \left(\mathbf{u}(t') \times \mathbf{a}(t') \right) \right]^2 \mathbf{e}_{r^*} \quad (4.56)$$

if we substitute

$$\mathbf{E}(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0} \frac{r^*}{\left(\mathbf{r}^* \cdot \mathbf{u}(t')\right)^3} \left[\mathbf{r}^* \times \left(\mathbf{u}(t') \times \mathbf{a}(t') \right) \right] \quad (4.57)$$

for $\mathbf{E}$ in (4.55).

Few more words about Poynting vector. Poynting vector as defined by (4.53) tells us about the power P of the EM radiation which crosses the unit area A in the direction to the normal of such area. Hence we write

$$dP = \mathbf{S}(\mathbf{r}, t) \cdot d\mathbf{A}. \quad (4.58)$$

Often, we are interested not in the power radiated per unit area, but in the power radiated per unit solid angle. Because, by definition of solid angle, an infinitesimally small area of a sphere dA subtends a solid angle $d\Omega = \frac{dA}{r^2}$, where r is the radius of a sphere, the equation (4.58) is often written as

$$dP = r^2 \mathbf{S}(\mathbf{r}, t) \cdot d\Omega. \quad (4.59)$$

In our case r^* is the radius of a sphere centered at the retarded position of the particle, thus the total power radiated by the charge is evaluated by substituting (4.56) into (4.59) and integrating over the entire sphere (all solid angles)

$$P(t) = c \frac{q^2}{16\pi^2\epsilon_0} \oint_{\Omega} \frac{r^{*4}}{\left(\mathbf{r}^* \cdot \mathbf{u}(t')\right)^6} \left[\mathbf{r}^* \times \left(\mathbf{u}(t') \times \mathbf{a}(t') \right) \right]^2 d\Omega. \quad (4.60)$$

4.2.1 Poynting vector for a point charge moving with non-relativistic velocities

Because we are discussing fields far away from the charge, until that moment we were using approximation where $\frac{1}{r^{*3}}$ and $\frac{1}{r^{*4}}$ terms in the Poynting vector were skipped as being significantly smaller comparing to $\frac{1}{r^{*2}}$ term. Let us still discuss far field zone, but introduce one more approximation. Let us consider charge, which is moving with speeds much slower than the speed of light ($v \ll c$). Then

$\mathbf{u}(t') = c\mathbf{e}_{r^*} - \mathbf{v}(t') \approx c\mathbf{e}_{r^*}$, $\mathbf{r}^* \cdot \mathbf{u}(t') = cr^*$ and the equation (4.60) can be written as

$$\begin{aligned} P(t) &= c \frac{q^2}{16\pi^2\epsilon_0} \oint_{\Omega} \frac{r^{*4}}{(cr^*)^6} \left[c\mathbf{e}_{r^*} (\mathbf{r}^* \cdot \mathbf{a}(t')) - \mathbf{a}(t') (\mathbf{r}^* \cdot c\mathbf{e}_{r^*}) \right]^2 d\Omega \\ &= \frac{q^2}{16\pi^2\epsilon_0 c^3} \oint_{\Omega} \frac{1}{r^{*2}} \left[\mathbf{e}_{r^*} (r^* a(t') \cos \Theta) - \mathbf{a}(t') r^* \right]^2 d\Omega \\ &= \frac{q^2}{16\pi^2\epsilon_0 c^3} \oint_{\Omega} a^2(t') \sin^2 \Theta d\Omega \quad (4.61) \end{aligned}$$

where Θ is the angle between direction of propagation $\mathbf{e}_{r^*}$ and the acceleration $\mathbf{a}(t')$ of the charged particle at the retarded time t' .

In spherical coordinates there is a formula for the differential $d\Omega$

$$d\Omega = \sin \Theta d\Theta d\phi. \quad (4.62)$$

Substituting (4.62) into (4.61) we get

$$P(t) = \frac{q^2 a^2(t')}{16\pi^2\epsilon_0 c^3} \int_0^\pi \sin^3 \Theta d\Theta \int_0^{2\pi} d\phi = \frac{q^2 a^2(t')}{16\pi^2\epsilon_0 c^3} \cdot \frac{4}{3} \cdot 2\pi = \frac{q^2 a^2(t')}{6\pi\epsilon_0 c^3}. \quad (4.63)$$

Equation (4.63) depicts the total radiation power evaluated at time t emitted from a non-relativistic accelerating particle at retarded time t' and is known as Larmor's formula.

4.2.2 Poynting vector for a point charge moving with relativistic velocities

Please note that radiation power P as defined in the previous section (equations (4.60), (4.61), and (4.63)) tells us about the rate at which energy crosses the sphere of radius r^* at time t , centered at the position of the particle at retarded time t' . The radiation leaves the charge at retarded time t' , after time interval $\frac{r^*}{c} = t - t'$ reaches the sphere and then (at time t) is integrated over the entire sphere to give total radiated power. That's why $P(t) = \frac{d\mathcal{E}}{dt}$ is the rate at which energy passes the sphere. If we want to calculate power which was emitted by the charge at the retarded time t' , we need to calculate $P(t') = \frac{d\mathcal{E}}{dt'}$. $P(t)$ and $P(t')$ are of course related by the following equations

$$P(t') = \frac{d\mathcal{E}}{dt'} = \frac{d\mathcal{E}}{dt} \frac{\partial t}{\partial t'} = P(t) \frac{\partial t}{\partial t'} \quad (4.64)$$

where

$$\frac{\partial t}{\partial t'} = \kappa = 1 - \frac{\mathbf{v}(t') \cdot \mathbf{e}_r^*}{c} = \frac{\mathbf{r}^* \cdot \mathbf{u}(t')}{r^* c}. \quad (4.65)$$

Hence

$$P(t') = \frac{q^2}{16\pi^2\epsilon_0} \oint_{\Omega} \frac{r^{*3}}{(\mathbf{r}^* \cdot \mathbf{u}(t'))^5} \left[\mathbf{r}^* \times (\mathbf{u}(t') \times \mathbf{a}(t')) \right]^2 d\Omega. \quad (4.66)$$

Integrating (4.66) over entire sphere (all solid angles $d\Omega$) will give us expression for total radiated power at retarded time t'

$$P(t') = \gamma^6 \frac{q^2}{6\pi\epsilon_0 c^3} \left[a^2(t') - \frac{1}{c^2} (\mathbf{v}(t') \times \mathbf{a}(t'))^2 \right]. \quad (4.67)$$

Equation (4.67) is a generalization of Larmor formula (4.63) for particles moving with relativistic velocities.

4.3 Visualizations of E, B and S fields

To make our considerations less tedious and to augment them with some drawings, I decided to write some small Python programs which will provide visualizations of electric and magnetic fields as well as Poynting vector for every discussed charge velocity and configuration. They are all available at <https://github.com/lukasz-wiecaszek/radiation-by-moving-charges>. In addition to static drawings, these programs also generate animations. So, if you would like to play with them, change some parameters, add more charges, generate your own animations or static drawing, feel free to download them and tweak according to your needs.

4.3.1 Point charge moving with non-relativistic velocities

Three figures below show the electric field, magnetic field and Poynting vector for a point charge moving on an elliptical trajectory. The velocity of the charge changes in the range from approximately $1.2 \cdot 10^6$ till $2.2 \cdot 10^6$ m/s, which is around 1 percent of the speed of light. The fields do not differ much from fields generated by a stationary charge.

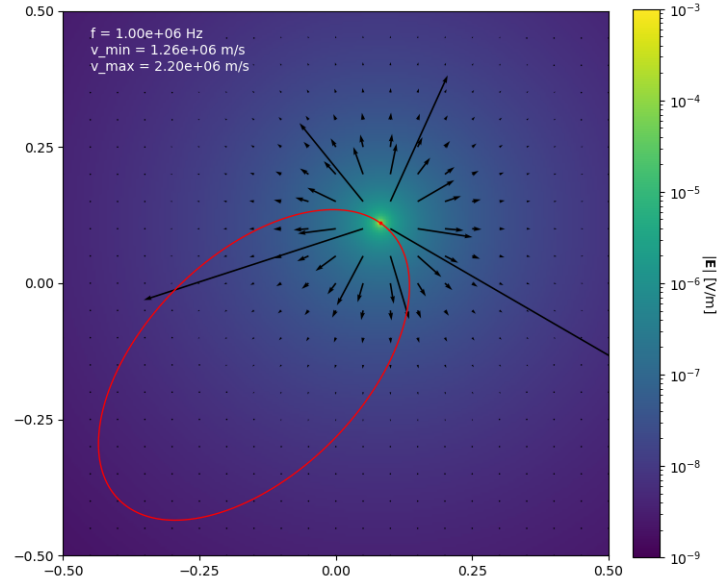


Figure 4.2: Electric field $\mathbf{E}$ of a point charge moving with arbitrary non-relativistic velocity.

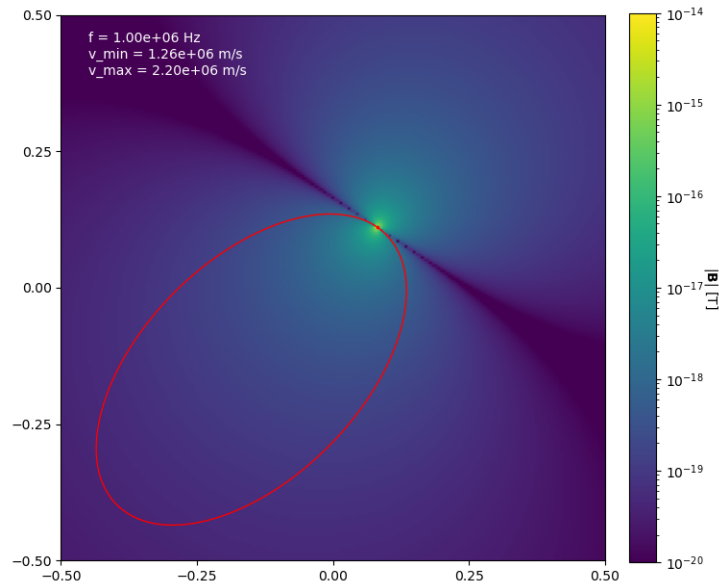


Figure 4.3: Magnetic field $\mathbf{B}$ of a point charge moving with arbitrary non-relativistic velocity.

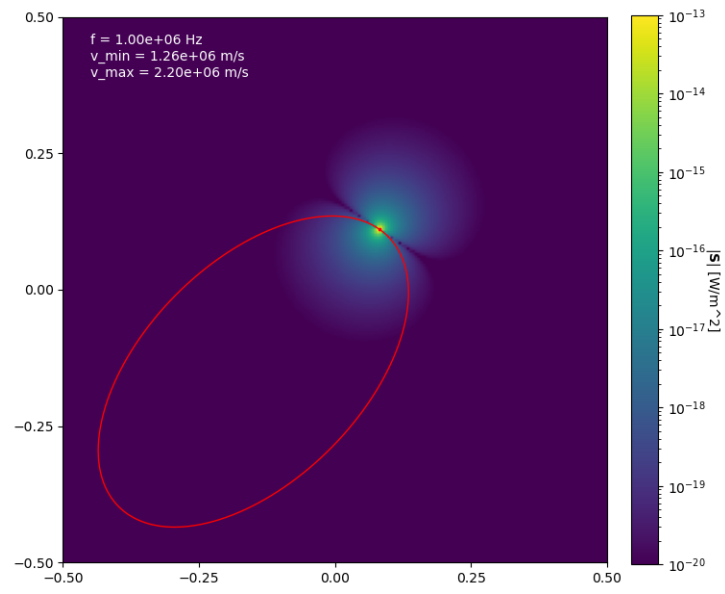


Figure 4.4: Poynting vector $\mathbf{S}$ of a point charge moving with arbitrary non-relativistic velocity.

4.3.2 Point charge moving with relativistic velocities

Here we have the same configuration as in the previous section, the charge is also moving on ellipsoidal trajectory, but this time with much higher velocities, reaching 88 % of the speed of light.

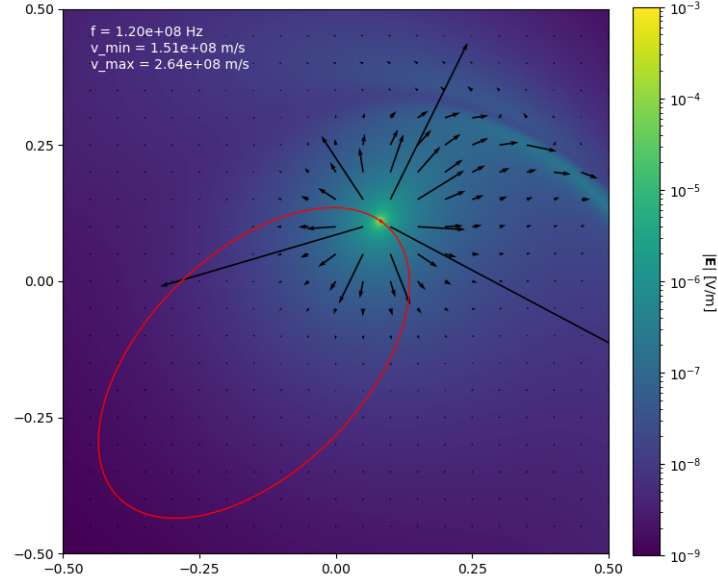


Figure 4.5: Electric field $\mathbf{E}$ of a point charge moving with arbitrary relativistic velocity.

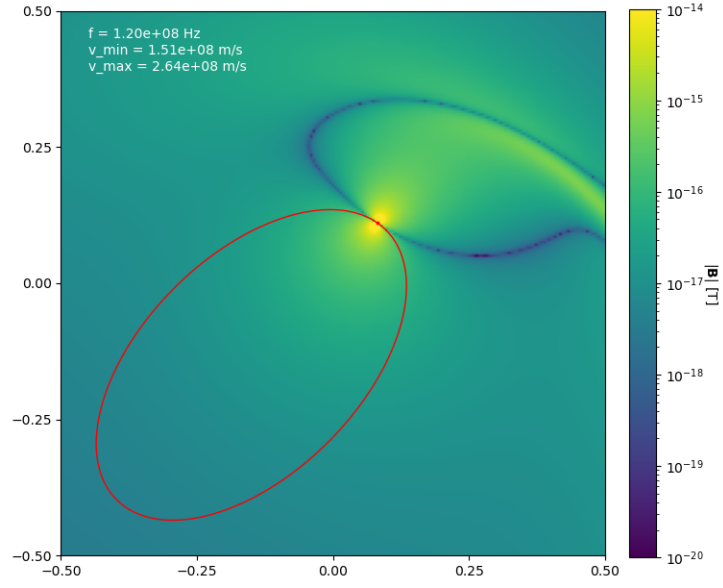


Figure 4.6: Magnetic field $\mathbf{B}$ of a point charge moving with arbitrary relativistic velocity.

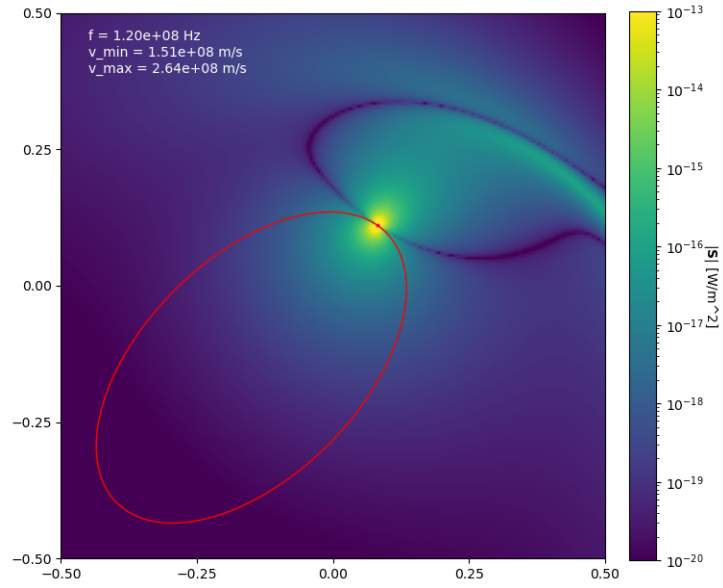


Figure 4.7: Poynting vector $\mathbf{S}$ of a point charge moving with arbitrary relativistic velocity.

5 Potentials and fields of a point charge moving with constant velocity

We already know that it is possible to derive electromagnetic fields (in closed form) of a point charge moving with arbitrary velocity, however equations (4.51) and (4.52) look a bit overwhelming. The question is, whether they reduce to some neater form if the charge was moving with a constant velocity in a straight line. Of course, one way to do so is to substitute $\mathbf{a} = 0$ and $\mathbf{v} = (v_x, 0, 0)$ in equation (4.51) and do required calculations. However, I will select another approach. I will start from the derivations of the Liénard–Wiechert potentials for a point charge moving with constant velocity and then calculate $\mathbf{E}$ field and $\mathbf{B}$ field from such obtained potentials. Let's begin then.

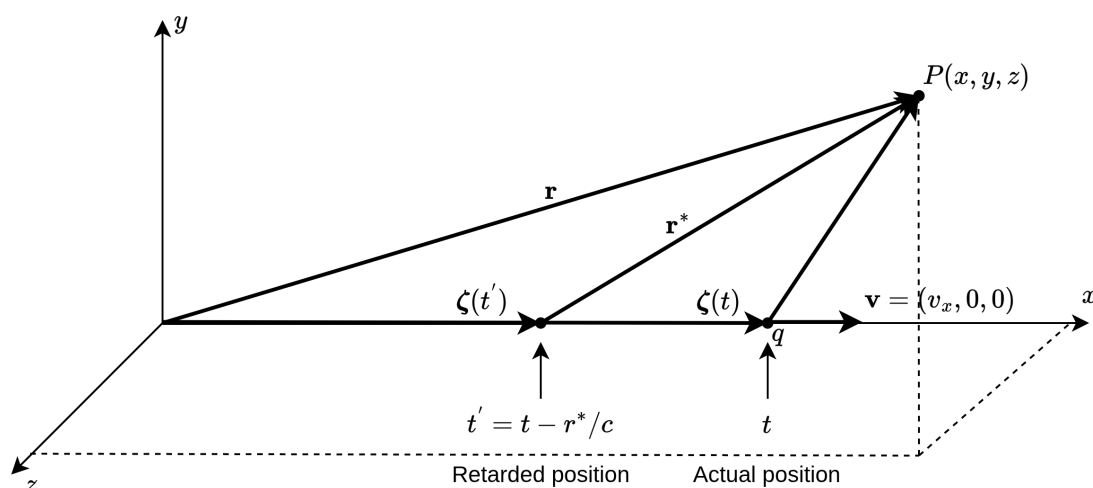


Figure 5.1: Point charge moving with constant velocity.

5.1 Liénard–Wiechert potentials

Suppose we have a charge moving with constant velocity $\mathbf{v} = (v_x, 0, 0)$ along the x -axis (see Figure 5.1). What we want to calculate are the Liénard–Wiechert potentials at the point $P(x, y, z)$ and time t . Let's assume that at time $t = 0$, the charge is at the origin. Hence, at time t the charge is at position $\boldsymbol{\zeta}(t) = (v_x t, 0, 0)$. But to evaluate potentials using formulas (3.37) and (3.38) we don't need to know the position at the time t , but the position at the retarded time

$$t' = t - \frac{r^*}{c}. \quad (5.1)$$

To calculate t' , we need to know

$$r^* = |\mathbf{r}^*| = \left| \mathbf{r} - \boldsymbol{\zeta}(t') \right| = \sqrt{(x - v_x t')^2 + y^2 + z^2}. \quad (5.2)$$

which on the other hand depends on t' . Thus we have the set of two equations (5.1) and (5.2) with two unknowns t' and r^* . Let us try to find t' first. To do so, we solve (5.1) for r^* , and substitute that into (5.2). After squaring both sides, we get

$$c^2 (t - t')^2 = (x - v_x t')^2 + y^2 + z^2. \quad (5.3)$$

Expanding the squared binomials leads to

$$(v_x^2 - c^2) t'^2 - 2(xv_x - c^2 t) t' + x^2 + y^2 + z^2 - (ct)^2 = 0. \quad (5.4)$$

This is a quadratic equation in t' which has two solutions

$$t'_\pm = \frac{xv_x - c^2 t \pm c \sqrt{(x - v_x t)^2 + \left(1 - \frac{v_x^2}{c^2}\right)(y^2 + z^2)}}{v_x^2 - c^2}. \quad (5.5)$$

Because retarded time t' must be smaller than t ($t' < t$), the only physically valid solution of (5.5) is the one with the positive sign. Then (5.5) can also be written as (numerator and denominator were divided by $-c^2$)

$$t' = \frac{t - \frac{xv_x}{c^2} - \frac{1}{c} \sqrt{(x - v_x t)^2 + \left(1 - \frac{v_x^2}{c^2}\right)(y^2 + z^2)}}{1 - \frac{v_x^2}{c^2}}. \quad (5.6)$$

Now, we may recall equation (3.37) for scalar potential $\phi(\mathbf{r}, t)$,

$$\phi(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0} \frac{1}{r^* - \frac{\mathbf{v}(t') \cdot \mathbf{r}^*}{c}}. \quad (5.7)$$

Since $\mathbf{v}$ is constant, the expression $\mathbf{v}(t') \cdot \mathbf{r}^*$ becomes

$$\mathbf{v}(t') \cdot \mathbf{r}^* = (v_x, 0, 0) \cdot (x - v_x t', 0, 0) = v_x (x - v_x t').$$

$r^* = c(t - t')$ and the expression $r^* - \frac{\mathbf{v}(t') \cdot \mathbf{r}^*}{c}$ evaluates to

$$r^* - \frac{\mathbf{v}(t') \cdot \mathbf{r}^*}{c} = c \left(t - t' \right) - \frac{v_x (x - v_x t')}{c} = c \left[t - \frac{v_x x}{c^2} - \left(1 - \frac{v_x^2}{c^2} \right) t' \right]. \quad (5.8)$$

Substituting t' from (5.6) we further get

$$r^* - \frac{\mathbf{v}(t') \cdot \mathbf{r}^*}{c} = \sqrt{(x - v_x t)^2 + \left(1 - \frac{v_x^2}{c^2} \right) (y^2 + z^2)},$$

and the whole scalar potential $\phi(\mathbf{r}, t)$ becomes

$$\phi(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0} \frac{1}{\sqrt{(x - v_x t)^2 + \left(1 - \frac{v_x^2}{c^2} \right) (y^2 + z^2)}} \quad (5.9)$$

or

$$\begin{aligned} \phi(\mathbf{r}, t) &= \frac{q}{4\pi\epsilon_0} \frac{1}{\sqrt{1 - \frac{v_x^2}{c^2}}} \frac{1}{\sqrt{\left(\frac{x - v_x t}{\sqrt{1 - v_x^2/c^2}} \right)^2 + y^2 + z^2}} = \\ &= \frac{q}{4\pi\epsilon_0} \frac{\gamma}{\sqrt{\gamma^2 (x - v_x t)^2 + y^2 + z^2}}, \end{aligned} \quad (5.10)$$

where $\gamma = 1/\sqrt{1 - v_x^2/c^2}$. To get expressions for vector potential $\mathbf{A}(\mathbf{r}, t)$ we need to augment scalar potential by the factor of $\mathbf{v}/c^2$. Doing so, vector potential reads

$$\mathbf{A}(\mathbf{r}, t) = \frac{\mathbf{v}}{c^2} \phi(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0 c^2} \frac{\mathbf{v}}{\sqrt{(x - v_x t)^2 + \left(1 - \frac{v_x^2}{c^2} \right) (y^2 + z^2)}} \quad (5.11)$$

or

$$\begin{aligned} \mathbf{A}(\mathbf{r}, t) &= \frac{q}{4\pi\epsilon_0 c^2} \frac{1}{\sqrt{1 - \frac{v_x^2}{c^2}}} \frac{\mathbf{v}}{\sqrt{\left(\frac{x - v_x t}{\sqrt{1 - v_x^2/c^2}} \right)^2 + y^2 + z^2}} = \\ &= \frac{q}{4\pi\epsilon_0 c^2} \frac{\gamma \mathbf{v}}{\sqrt{\gamma^2 (x - v_x t)^2 + y^2 + z^2}}. \end{aligned} \quad (5.12)$$

Equations (5.9), (5.10), (5.11) and (5.12) are the expressions for Liénard–Wiechert potentials of a point charge moving with constant velocity v_x .

5.2 Electric and magnetic fields

Once we know the potentials, we can compute the electric and magnetic fields using already well known relations:

$$\mathbf{E} = -\nabla\phi - \frac{\partial\mathbf{A}}{\partial t}, \quad (5.13)$$

$$\mathbf{B} = \nabla \times \mathbf{A}. \quad (5.14)$$

Let's start with the evaluation of $\nabla\phi$.

$$\begin{aligned} \frac{\partial\phi}{\partial x} &= -\gamma^3 \frac{q}{4\pi\epsilon_0} \frac{x - v_x t}{[\gamma^2 (x - v_x t)^2 + y^2 + z^2]^{3/2}} \\ \frac{\partial\phi}{\partial y} &= -\gamma \frac{q}{4\pi\epsilon_0} \frac{y}{[\gamma^2 (x - v_x t)^2 + y^2 + z^2]^{3/2}} \\ \frac{\partial\phi}{\partial z} &= -\gamma \frac{q}{4\pi\epsilon_0} \frac{z}{[\gamma^2 (x - v_x t)^2 + y^2 + z^2]^{3/2}} \end{aligned} \quad (5.15)$$

The second term, $\frac{\partial\mathbf{A}}{\partial t}$ gives

$$\frac{\partial\mathbf{A}}{\partial t} = \gamma^3 \frac{q_1}{4\pi\epsilon_0} \frac{(x - vt)v \frac{\mathbf{v}}{c^2}}{[\gamma^2 (x - vt)^2 + y^2 + z^2]^{3/2}} \quad (5.16)$$

Adding those two terms together ($\nabla\phi$ and $\frac{\partial\mathbf{A}}{\partial t}$) and noting that $\mathbf{v} = (v_x, 0, 0)$, the components of the electric field $\mathbf{E}$ gain the following form:

$$\begin{aligned} E_x &= \gamma^3 \frac{q}{4\pi\epsilon_0} \frac{(x - v_x t)}{[\gamma^2 (x - v_x t)^2 + y^2 + z^2]^{3/2}} - \gamma^3 \frac{q}{4\pi\epsilon_0} \frac{(x - v_x t) \frac{v_x^2}{c^2}}{[\gamma^2 (x - v_x t)^2 + y^2 + z^2]^{3/2}} = \\ &= \gamma^3 \frac{q}{4\pi\epsilon_0} \frac{(x - v_x t)(1 - \frac{v_x^2}{c^2})}{[\gamma^2 (x - v_x t)^2 + y^2 + z^2]^{3/2}} = \gamma \frac{q}{4\pi\epsilon_0} \frac{(x - v_x t)}{[\gamma^2 (x - v_x t)^2 + y^2 + z^2]^{3/2}}, \end{aligned}$$

$$E_y = \gamma \frac{q}{4\pi\epsilon_0} \frac{y}{[\gamma^2 (x - v_x t)^2 + y^2 + z^2]^{3/2}},$$

$$E_z = \gamma \frac{q}{4\pi\epsilon_0} \frac{z}{[\gamma^2 (x - v_x t)^2 + y^2 + z^2]^{3/2}}.$$

To calculate $\mathbf{B}$, let us recall yet another vector calculus identity

$$\nabla \times (s\mathbf{u}) = (\nabla s) \times \mathbf{u} + s(\nabla \times \mathbf{u}), \quad (5.17)$$

where s is an arbitrary scalar and $\mathbf{u}$ arbitrary vector field. Then

$$\mathbf{B} = \nabla \times \mathbf{A} = \nabla \times \left(\frac{\mathbf{v}}{c^2} \phi \right) = (\nabla \phi) \times \frac{\mathbf{v}}{c^2} + \phi \left(\nabla \times \frac{\mathbf{v}}{c^2} \right).$$

Because $\mathbf{v}$ is constant, $\nabla \times \mathbf{v} = 0$, and because of that, also $\mathbf{A}$, $\frac{\partial \mathbf{A}}{\partial t}$ and $\mathbf{v}$ are collinear, so $\mathbf{v} \times \frac{\partial \mathbf{A}}{\partial t}$. Taking those facts into considerations, the whole expression for $\mathbf{B}$ evaluates to

$$\mathbf{B} = -\frac{\mathbf{v}}{c^2} \times \nabla \phi = \frac{\mathbf{v}}{c^2} \times \left(\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t} \right) = \frac{\mathbf{v}}{c^2} \times \mathbf{E}. \quad (5.18)$$

Thus, for a point charge moving with constant velocity $\mathbf{v} = (v_x, 0, 0)$, the explicit expressions for the fields at point $P(x, y, z)$ and time t are:

$$\mathbf{E}(\mathbf{r}, t) = \gamma \frac{q}{4\pi\epsilon_0} \frac{\mathbf{r} - \mathbf{v}t}{[\gamma^2 (x - v_x t)^2 + y^2 + z^2]^{3/2}} \quad (5.19)$$

$$\mathbf{B}(\mathbf{r}, t) = \frac{1}{c^2} \mathbf{v} \times \mathbf{E}(\mathbf{r}, t). \quad (5.20)$$

These fields describe the electromagnetic field of a point charge moving with constant velocity, and reduce to the familiar Coulomb field in the limit $v_x \rightarrow 0$. And this is indeed remarkable result. Despite the fact that the contribution to the electric and magnetic fields comes from the retarded time t' , the direction of the fields itself is always determined by the instantaneous position of the charge, i.e., by the vector $\mathbf{r} - \mathbf{v}t$. This means that, although the magnitude of the fields depends on the retarded position, their direction at any given point and time is simply along the line joining the current position of the charge and the point of observation (see Figure 5.5). The field is "amplified" (by a factor $\frac{1}{\sqrt{1 - v_x^2/c^2}}$) in the direction perpendicular to $\mathbf{v}$ and "attenuated" (by a factor $1 - v_x^2/c^2$) in directions parallel to $\mathbf{v}$.

5.3 The Poynting vector

The Poynting vector for charge moving with constant velocity is

$$\mathbf{S}(\mathbf{r}, t) = c^2 \epsilon_0 \mathbf{E}(\mathbf{r}, t) \times \mathbf{B}(\mathbf{r}, t) = \epsilon_0 \mathbf{E} \times (\mathbf{v} \times \mathbf{E}) = \frac{\gamma^2 q^2}{16\pi^2 \epsilon_0} \frac{(\mathbf{r} - \mathbf{v}t) \times (\mathbf{v} \times \mathbf{r})}{[\gamma^2 (x - v_x t)^2 + y^2 + z^2]^3} \approx \frac{1}{r^4}. \quad (5.21)$$

There is no acceleration thus there is no radiation term. But how to interpret the term which is still present in the expression (5.21). As Poynting vector is nonzero, there is still flow of the energy flux (per unit time). It appears that in the case of uniformly moving charges Poynting vector describes a local flow of the bound (non-radiative) field energy that is carried along with the charge. Generally speaking, regardless whether the charge is accelerating or not, the $1/r^4$ term in the Poynting vector represents redistribution and transport of bound field energy and momentum, not energy radiated away to infinity.

5.4 Visualizations of E, B and S fields

5.4.1 Point charge moving with non-relativistic velocities

Three figures below show the electric field, magnetic field and Poynting vector for a point charge moving along the straight line (x-axis, or from left to right) with constant velocity $v = 0.01c$. The fields does not differ much from fields generated by a stationary charge.

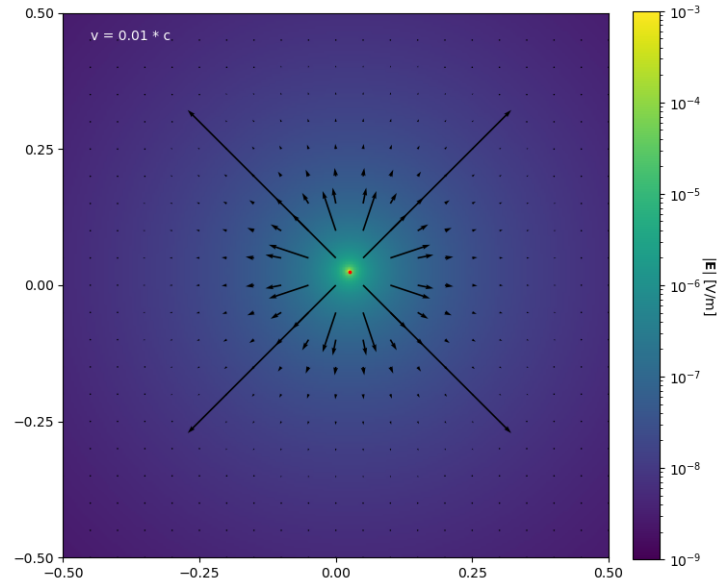


Figure 5.2: Electric field $\mathbf{E}$ of a point charge moving with constant non-relativistic velocity.

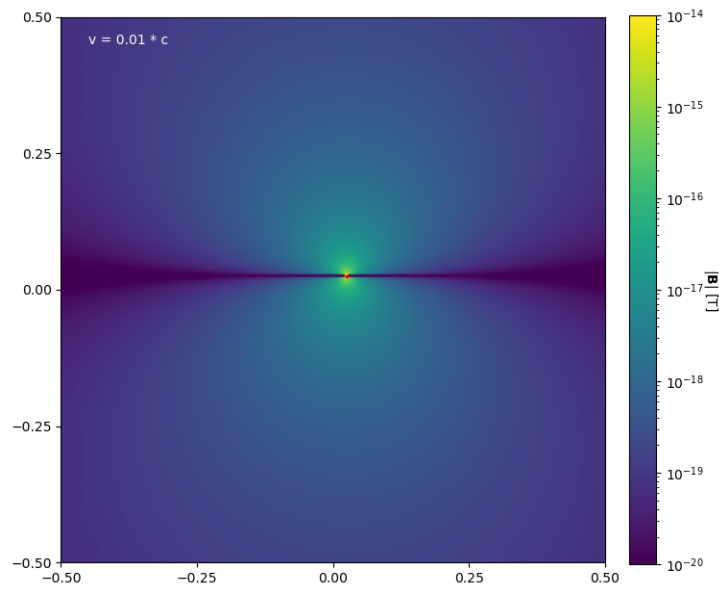


Figure 5.3: Magnetic field $\mathbf{B}$ of a point charge moving with constant non-relativistic velocity.

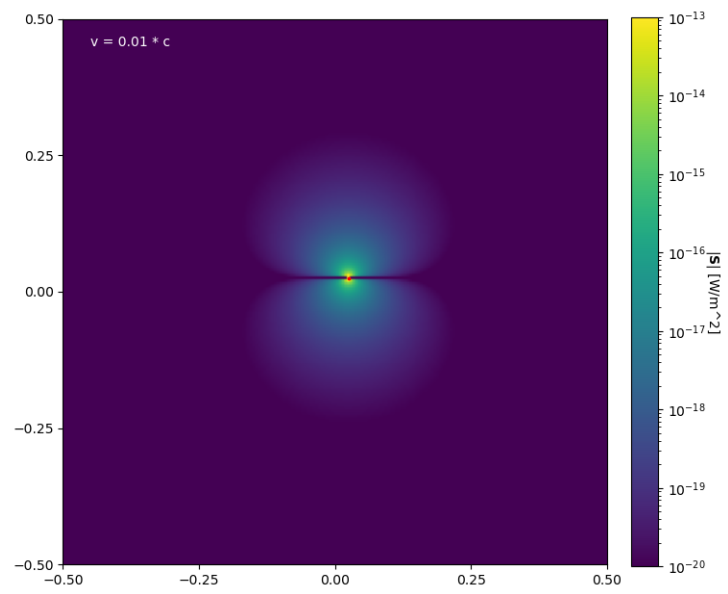


Figure 5.4: Poynting vector $\mathbf{S}$ of a point charge moving with constant non-relativistic velocity.

5.4.2 Point charge moving with relativistic velocities

Here we have the same configuration as in the previous section, the charge is also moving along the straight line with constant velocity, but this time magnitude of the velocity is much higher and equals 99 % of the speed of light ($v = 0.99c$).

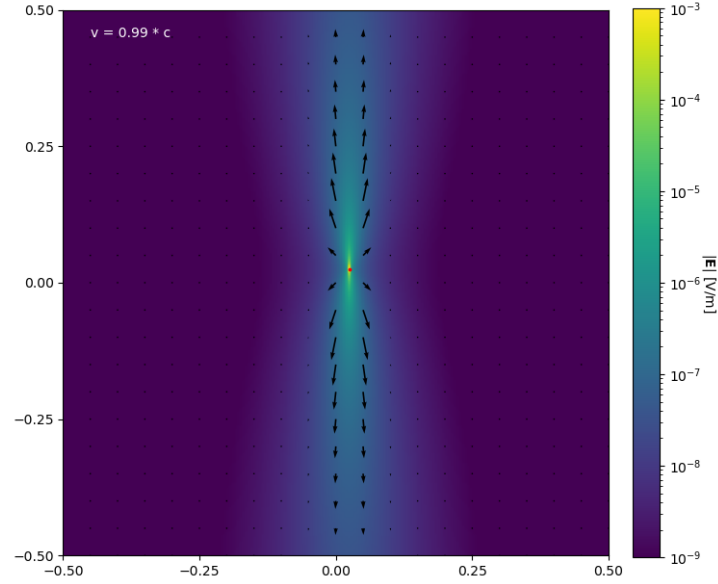


Figure 5.5: Electric field $\mathbf{E}$ of a point charge moving with constant relativistic velocity.

It is now clearly visible, that in case of relativistic speeds, the fields are flattened out in the direction perpendicular to the motion.

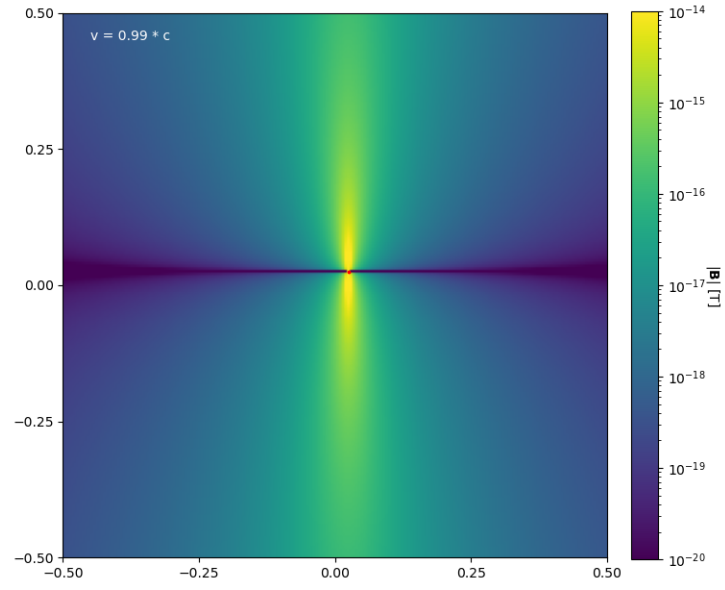


Figure 5.6: Magnetic field $\mathbf{B}$ of a point charge moving with constant relativistic velocity.

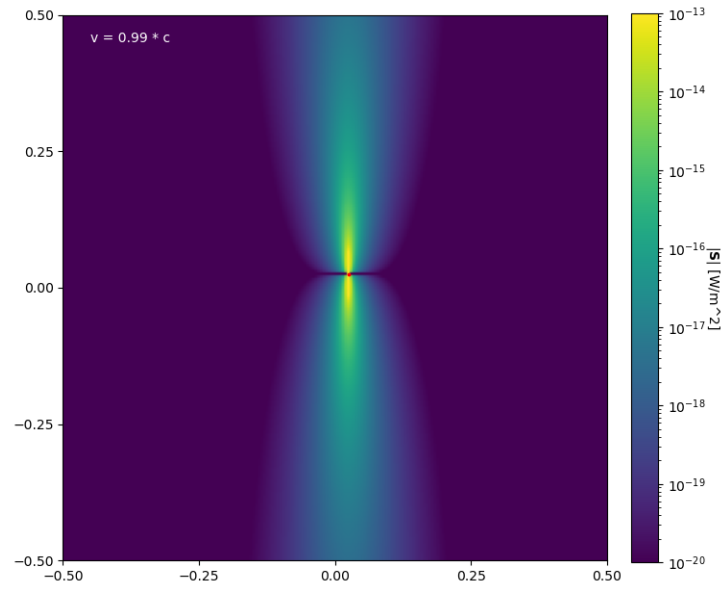


Figure 5.7: Poynting vector $\mathbf{S}$ of a point charge moving with constant relativistic velocity.

6 Radiation from an oscillating single charge

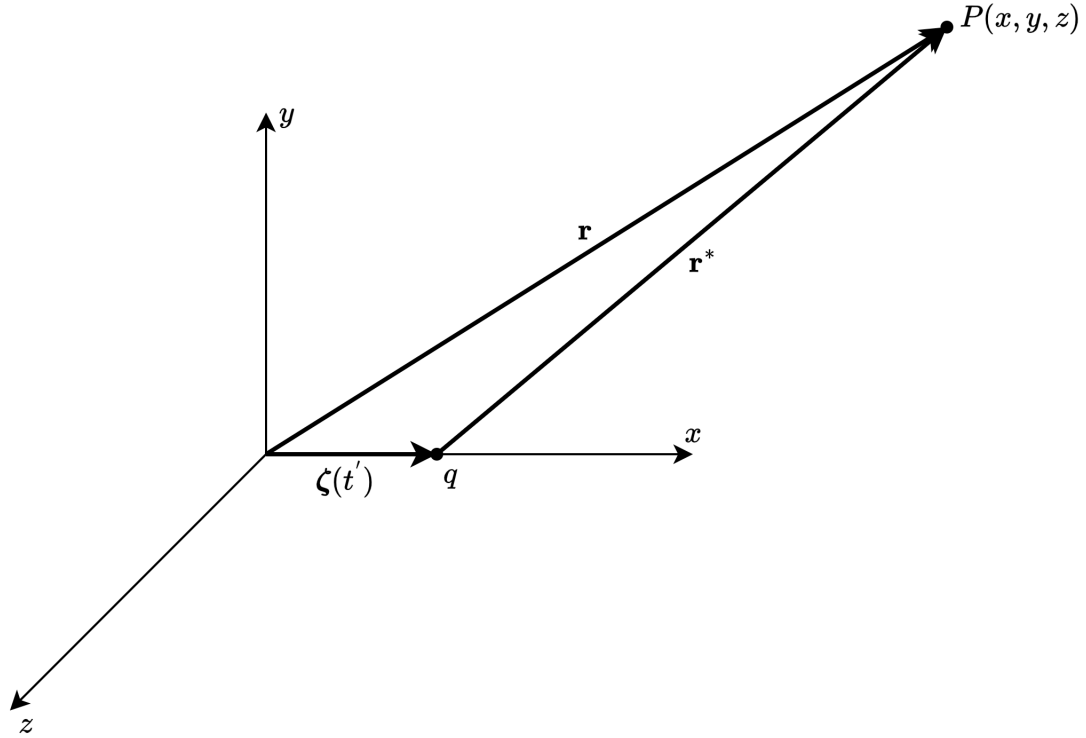


Figure 6.1: Oscillating single charge.

6.1 Radiated power

Let us consider the case, where the point charge q oscillates along the x -axis and the oscillations are described by the cosinusoidal function of time

$$x(t) = \zeta(t) = A \cos(\omega t), \quad (6.1)$$

where A is the amplitude, and $\omega = 2\pi f$ is the angular frequency of the oscillations. Then

$$v(t) = -A\omega \sin(\omega t), \quad (6.2)$$

and

$$a(t) = -A\omega^2 \cos(\omega t). \quad (6.3)$$

Let us now find the radiated power from such oscillating single charge for two scenarios. First, where the charge is moving with non-relativistic velocity, and the second time, when the charge is moving with relativistic velocities.

6.1.1 Non-relativistic case

We will start this discussion by recalling the Larmor formula (equation 4.63) from chapter 4:

$$P(t) = \frac{d\mathcal{E}}{dt} = -\frac{q^2 a^2(t')}{6\pi\epsilon_0 c^3}. \quad (6.4)$$

Larmor formula is exactly what we need here. It describes the total radiation power evaluated at time t emitted from a non-relativistic accelerating particle at retarded time t' .

Substituting $a(t)$ into (6.4) we get the expression for the instantaneous loss of energy (radiated power) of a single charge due to its oscillation. This is

$$P(t) = \frac{d\mathcal{E}}{dt} = -\frac{q^2}{6\pi\epsilon_0 c^3} A^2 \omega^4 \cos^2(\omega t'). \quad (6.5)$$

From the equation (6.5) it follows that the radiated power varies with time according to the square of the cosine. The time-average value of the cosine is $1/2$. Therefore, the time-averaged radiated power is equal to

$$\langle P \rangle = -\frac{1}{2} \frac{q^2}{6\pi\epsilon_0 c^3} A^2 \omega^4. \quad (6.6)$$

Hence, a single charge oscillating with non-relativistic velocity radiates power proportional to

$$\langle P \rangle \propto A^2 \omega^4. \quad (6.7)$$

6.1.2 Relativistic case

Let me now repeat the same derivations, but this time for a charge moving with relativistic velocity. So we have to start our considerations from a relativistic version of the Larmor formula. This is equation (4.67) from chapter 4:

$$P(t') = -\gamma^6 \frac{q^2}{6\pi\epsilon_0 c^3} \left[a^2(t') - \frac{1}{c^2} \left(\mathbf{v}(t') \times \mathbf{a}(t') \right)^2 \right], \quad (6.8)$$

where $P(t') = \frac{d\mathcal{E}}{dt'}$ is the instantaneous power radiated by a charged particle moving with a relativistic velocity.

In the case of an oscillating charge, the $\mathbf{v}(t')$ is all the time parallel to the $\mathbf{a}(t')$, so that $\mathbf{v}(t') \times \mathbf{a}(t') = 0$ and the equation (4.67) takes the following form:

$$P(t') = -\gamma^6 \frac{q^2}{6\pi\epsilon_0 c^3} a^2(t'). \quad (6.9)$$

Substituting $a(t)$ into (6.9) we get the expression for the instantaneous loss of energy (radiated power) of a charge oscillating with relativistic velocity. This is

$$P(t') = \frac{d\mathcal{E}}{dt'} = -\gamma^6 \frac{q^2}{6\pi\epsilon_0 c^3} A^2 \omega^4 \sin^2(\omega t), \quad (6.10)$$

and the time-averaged radiated power this time is equal to

$$\langle P \rangle = -\frac{1}{2} \gamma^6 \frac{q^2}{6\pi\epsilon_0 c^3} A^2 \omega^4. \quad (6.11)$$

Hence, a single charge oscillating with relativistic velocity radiates power proportional to

$$\langle P \rangle \propto \gamma^6 A^2 \omega^4, \quad (6.12)$$

which differs from the non-relativistic case by the factor of γ^6 . This means that as the charge velocity approaches the speed of light, the radiated power increases by the sixth power of the Lorentz factor, making relativistic effects extremely significant at high frequencies or amplitudes.

This strong dependence on γ^6 explains why highly relativistic particles are extremely efficient radiators. For instance, in synchrotron facilities and astrophysical sources, particles moving at relativistic speeds can radiate enormous amounts of energy, which must be accounted for in the dynamics of the system. The γ^6 enhancement is one of the most important reasons why particle accelerators must supply increasingly large amounts of energy to maintain the motion of relativistic particles.

6.2 Visualizations of E, B and S fields

6.2.1 Point charge moving with non-relativistic velocities

The following three graphs show the electric and magnetic fields as well as the Poynting vector field for a point charge oscillating along the x-axis, where oscillations are described by the sinusoidal function of time ($x(t) = \zeta(t) = A \sin(2\pi f t)$). With the $f = 1.00 \times 10^8$ Hz, the maximum speed that the charge reaches is equal to 3.14×10^7 m/s, which is around 10 % of the speed of light. The wavelength of the radiation at that frequency is $\lambda = c/f \approx 3$ m, which at that scale is not visible in the attached graphs.

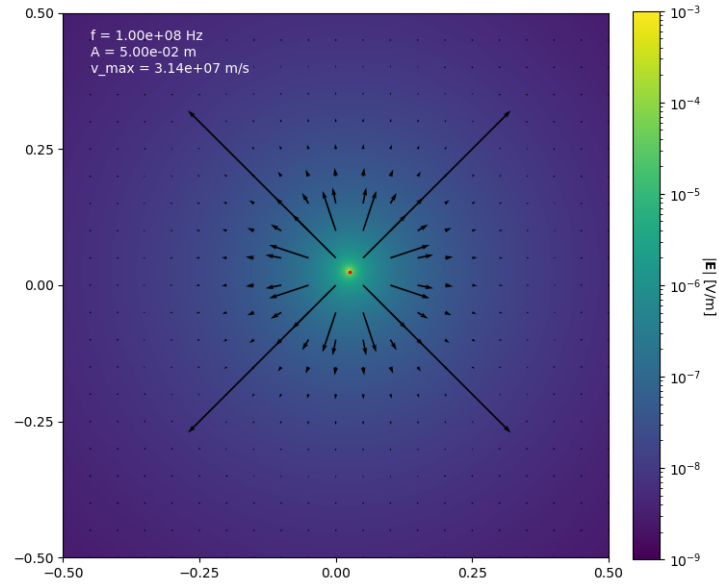


Figure 6.2: Electric field $\mathbf{E}$ of a single oscillating point charge moving with non-relativistic velocity.

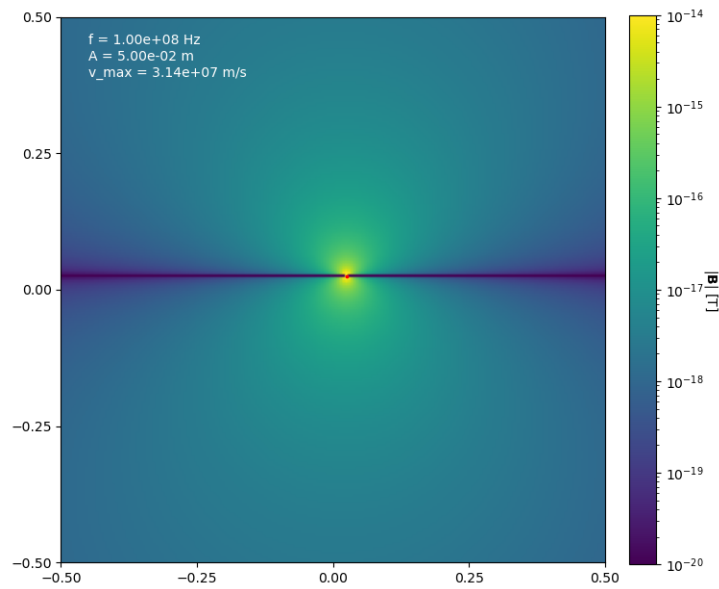


Figure 6.3: Magnetic field $\mathbf{B}$ of a single oscillating point charge moving with non-relativistic velocity.

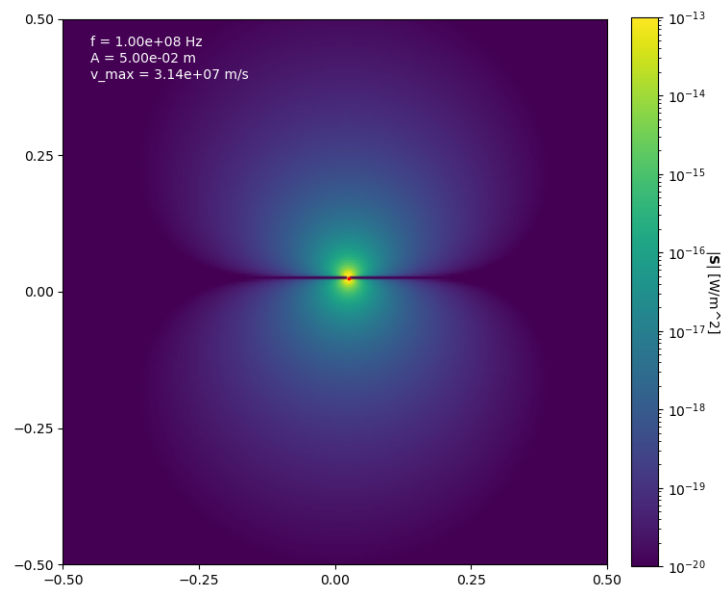


Figure 6.4: Poynting vector $\mathbf{S}$ of a single oscillating point charge moving with non-relativistic velocity.

6.2.2 Point charge moving with relativistic velocities

But if we increase the frequency from 1.00×10^8 Hz to 9.00×10^8 Hz, the wavelength will reduce to $\lambda \approx 0.33$ m, and that wavelength with its crests and troughs will now be visible on the following figures. Please also note, that with the frequency of 9.00×10^8 Hz the maximum speed the particle reaches is 2.83×10^8 m/s, which is around 94 % of the speed of light.

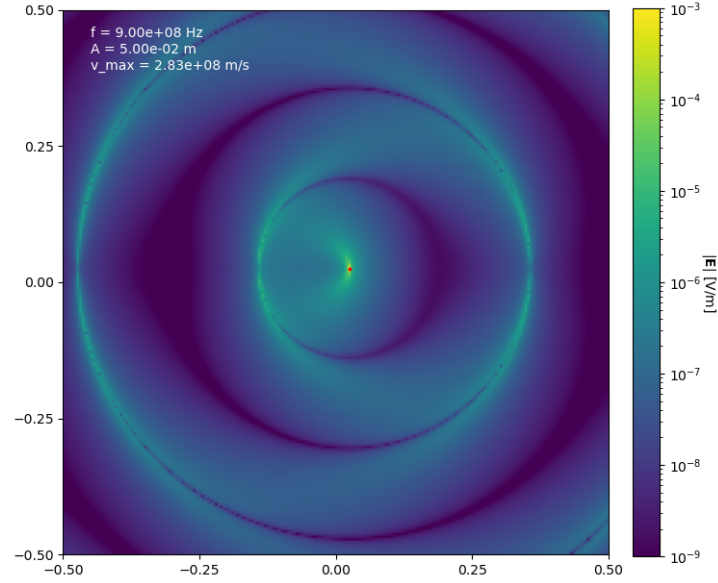


Figure 6.5: Electric field $\mathbf{E}$ of a single oscillating point charge moving with relativistic velocity.

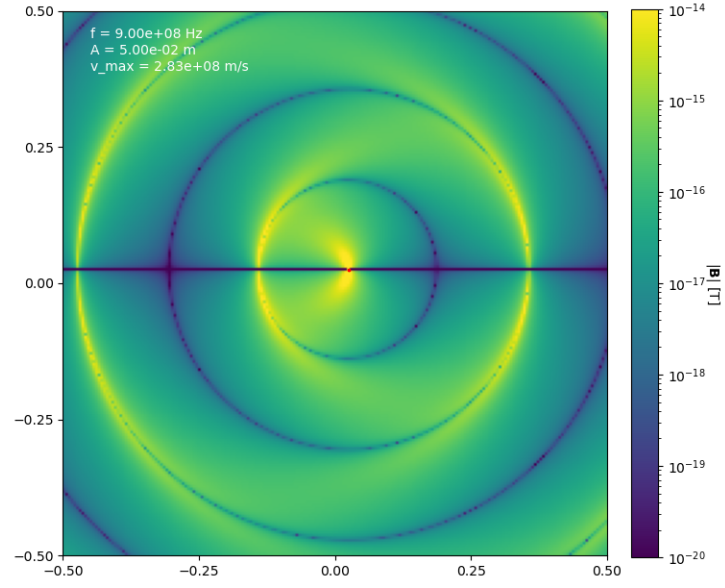


Figure 6.6: Magnetic field $\mathbf{B}$ of a single oscillating point charge moving with relativistic velocity.

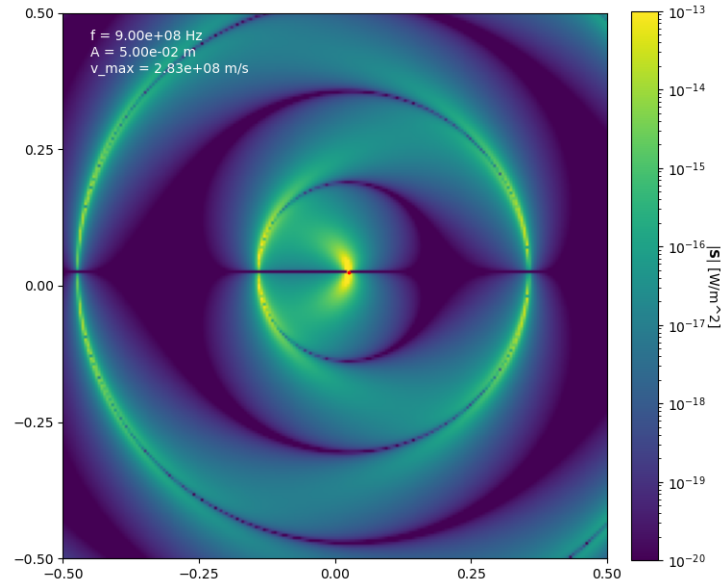


Figure 6.7: Poynting vector $\mathbf{S}$ of a single oscillating point charge moving with relativistic velocity.

7 Radiation from an oscillating electric dipole

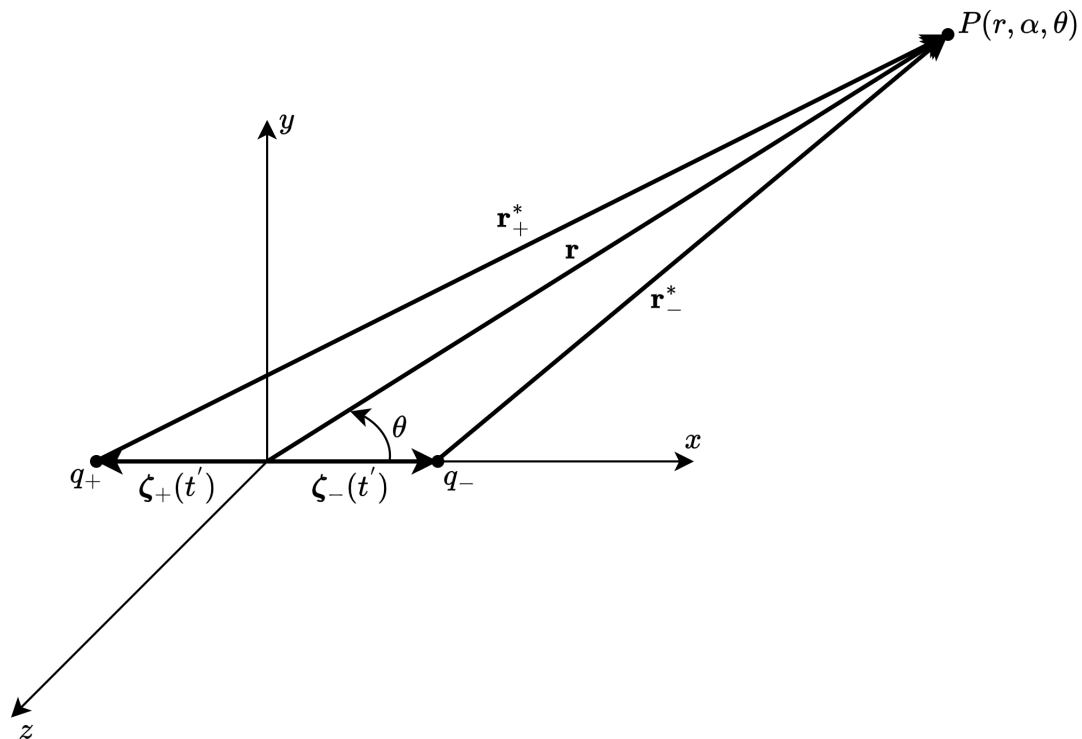


Figure 7.1: Oscillating electric dipole.

In the previous chapter, I derived the equation for the instantaneous power radiated by a charged particle oscillating along the x-axis, where the oscillations were described by the cosinusoidal function of time

$$\zeta(t) = A \cos(\omega t) \mathbf{e}_x. \quad (7.1)$$

Let me now extend this scenario and add one more particle with the same charge in absolute value, but opposite sign, with the oscillations in the opposite direction to the first particle. That way we have a system with two oscillating particles:

$$\zeta_-(t) = A \cos(\omega t) \mathbf{e}_x, \quad (7.2)$$

$$\zeta_+(t) = -A \cos(\omega t) \mathbf{e}_x. \quad (7.3)$$

But such a system is nothing more than just an oscillating electric dipole:

$$\mathbf{p}(t) = q(\boldsymbol{\zeta}_-(t) - \boldsymbol{\zeta}_+(t)) = q2A \cos(\omega t) \mathbf{e}_x = p_0 \cos(\omega t) \mathbf{e}_x, \quad (7.4)$$

where $p_0 = qd$ ($d = 2A$) is the maximum value of the electric dipole moment, $\omega = 2\pi f$ is the angular frequency of the oscillations, and $\mathbf{e}_x$ is the x-axis' unit vector.

Please note that the same model of an oscillating electric dipole can in fact describe two different scenarios. One (used above), where we have two point charges oscillating along a common axis (q constant, d oscillates), and the other, where let's say two tiny metal spheres are separated by a distance d and connected by a wire to a power supply which moves charges between the spheres according to the following formula (d constant, q oscillates):

$$q(t) = q_0 \cos(\omega t). \quad (7.5)$$

We could say that the radiated power from such a configuration is just twice as big as from the single oscillating charge and thus we can end further considerations. However, let me approach this problem in a similar fashion as it was done in chapter 5, and start from derivations of the scalar and vector potentials of an oscillating electric dipole, then calculate $\mathbf{E}$ and $\mathbf{B}$ fields, and finally $\mathbf{S}$ field and the radiated power. Let's begin then.

7.1 Liénard–Wiechert potentials

We will start this section from recalling expressions for scalar $\phi(\mathbf{r}, t)$, and vector $\mathbf{A}(\mathbf{r}, t)$ Liénard–Wiechert potentials of a moving point charge. These are:

$$\phi(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0} \frac{1}{\kappa r^*}, \quad (7.6)$$

$$\mathbf{A}(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0 c^2} \frac{\mathbf{v}(t')}{\kappa r^*}, \quad (7.7)$$

where $\kappa = 1 - \frac{\mathbf{v}(t') \cdot \mathbf{e}_{r^*}}{c}$, and $\mathbf{v}(t') = \frac{d\boldsymbol{\zeta}(t')}{dt'}$.

Therefore, the overall potential of an electric dipole at the point of observation $\mathbf{r}$ and time t is the sum of the potentials from the individual charges. In our case (see figure 7.1), this is:

$$\begin{aligned}\phi(\mathbf{r}, t) = \phi_+(\mathbf{r}, t) + \phi_-(\mathbf{r}, t) &= \frac{+q}{4\pi\epsilon_0} \frac{1}{\kappa_+ r_+^*} + \frac{-q}{4\pi\epsilon_0} \frac{1}{\kappa_- r_-^*} \\ &= \frac{q}{4\pi\epsilon_0} \left(\frac{1}{\kappa_+ r_+^*} - \frac{1}{\kappa_- r_-^*} \right), \quad (7.8)\end{aligned}$$

$$\begin{aligned}\mathbf{A}(\mathbf{r}, t) = \mathbf{A}_+(\mathbf{r}, t) + \mathbf{A}_-(\mathbf{r}, t) &= \frac{+q}{4\pi\epsilon_0 c^2} \frac{-\mathbf{v}(t'_+)}{\kappa_+ r_+^*} + \frac{-q}{4\pi\epsilon_0 c^2} \frac{+\mathbf{v}(t'_-)}{\kappa_- r_-^*} \\ &= -\frac{q}{4\pi\epsilon_0 c^2} \left(\frac{\mathbf{v}(t'_+)}{\kappa_+ r_+^*} + \frac{\mathbf{v}(t'_-)}{\kappa_- r_-^*} \right). \quad (7.9)\end{aligned}$$

At this moment, let me note, that as long as the $\zeta(t') < r$, the expression

$$\frac{1}{r^*} = \frac{1}{|\mathbf{r} - \boldsymbol{\zeta}(t')|} \quad (7.10)$$

can be written using Legendre polynomials as

$$\frac{1}{r^*} = \frac{1}{|\mathbf{r} - \boldsymbol{\zeta}(t')|} = \frac{1}{r} \sum_{n=0}^{\infty} \left(\frac{\zeta(t')}{r} \right)^n P_n(\cos \theta), \quad (7.11)$$

where $P_n(\cos \theta)$ is the n^{th} Legendre polynomial.

Thus

$$\frac{1}{\kappa_+ r_+^*} = \frac{1}{\kappa_+ r} \sum_{n=0}^{\infty} (-1)^n \left(\frac{A \cos(\omega t'_+)}{r} \right)^n P_n(\cos \theta), \quad (7.12)$$

and

$$\frac{1}{\kappa_- r_-^*} = \frac{1}{\kappa_- r} \sum_{n=0}^{\infty} \left(\frac{A \cos(\omega t'_-)}{r} \right)^n P_n(\cos \theta). \quad (7.13)$$

Then

$$\begin{aligned}\phi(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0 r} &\left[\frac{1}{\kappa_+} \sum_{n=0}^{\infty} (-1)^n \left(\frac{A \cos(\omega t'_+)}{r} \right)^n P_n(\cos \theta) \right. \\ &\left. - \frac{1}{\kappa_-} \sum_{n=0}^{\infty} \left(\frac{A \cos(\omega t'_-)}{r} \right)^n P_n(\cos \theta) \right], \quad (7.14)\end{aligned}$$

$$\mathbf{A}(\mathbf{r}, t) = -\frac{q}{4\pi\epsilon_0 c^2 r} \left[\frac{\mathbf{v}(t'_+)}{\kappa_+} \sum_{n=0}^{\infty} (-1)^n \left(\frac{A \cos(\omega t'_+)}{r} \right)^n P_n(\cos \theta) + \frac{\mathbf{v}(t'_-)}{\kappa_-} \sum_{n=0}^{\infty} \left(\frac{A \cos(\omega t'_-)}{r} \right)^n P_n(\cos \theta) \right]. \quad (7.15)$$

At this stage, let me introduce one of the few approximations which will make our derivations much easier, and on the other hand will not introduce significant error to the presented solution. If the extension of the dipole d is much smaller than the distance between the dipole and the point of observation $\mathbf{r}$, then

$$t'_+ \simeq t'_- \simeq t' = t - \frac{r}{c}. \quad (7.16)$$

Armed with that knowledge, and expanding only first two terms from the infinite sums, we further get

$$\phi(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0 r} \left[\left(\frac{1}{\kappa_+} - \frac{1}{\kappa_-} \right) - \left(\frac{1}{\kappa_+} + \frac{1}{\kappa_-} \right) \left(\frac{A \cos(\omega t')}{r} \right)^1 P_1(\cos \theta) \right], \quad (7.17)$$

$$\mathbf{A}(\mathbf{r}, t) = -\frac{q\mathbf{v}(t')}{4\pi\epsilon_0 c^2 r} \left[\left(\frac{1}{\kappa_+} + \frac{1}{\kappa_-} \right) + \left(\frac{1}{\kappa_-} - \frac{1}{\kappa_+} \right) \left(\frac{A \cos(\omega t')}{r} \right)^1 P_1(\cos \theta) \right]. \quad (7.18)$$

Higher terms are out of our scope as they are proportional to $1/r^3$, $1/r^4$, and so on. They give rise to an electric field which is vanishingly small at large distances from the dipole (in the far zone).

Now, let me recall that

$$\begin{aligned} \kappa_+ &= 1 - \frac{1}{c} \frac{d\boldsymbol{\zeta}_+(t'_+)}{dt'} \cdot \mathbf{e}_{r_+}^* = 1 - \frac{1}{c} \frac{d(-A \cos(\omega t'_+) \mathbf{e}_x)}{dt'} \cdot \mathbf{e}_{r_+}^* \\ &= 1 - \frac{w}{c} A \sin(\omega t'_+) \mathbf{e}_x \cdot \mathbf{e}_{r_+}^* \end{aligned} \quad (7.19)$$

and

$$\begin{aligned}\kappa_- &= 1 - \frac{1}{c} \frac{d\zeta_-(t'_-)}{dt'} \cdot \mathbf{e}_{r_-^*} = 1 - \frac{1}{c} \frac{d\left(A \cos(\omega t'_-) \mathbf{e}_x\right)}{dt'} \cdot \mathbf{e}_{r_-^*} \\ &= 1 + \frac{w}{c} A \sin(\omega t'_-) \mathbf{e}_x \cdot \mathbf{e}_{r_-^*}. \quad (7.20)\end{aligned}$$

Because we are still discussing far fields ($d \ll r$), our approximation $t'_+ \simeq t'_- \simeq t'$ automatically imposes that $\mathbf{e}_{r_+^*} \simeq \mathbf{e}_{r_-^*} \simeq \mathbf{e}_{r^*}$. Then, applying approximations and noting that $\mathbf{e}_x \cdot \mathbf{e}_{r^*} = \cos \theta$, we obtain

$$\kappa_+ = 1 - \frac{w}{c} A \sin(\omega t') \cos \theta, \quad (7.21)$$

$$\kappa_- = 1 + \frac{w}{c} A \sin(\omega t') \cos \theta. \quad (7.22)$$

Once we know the expression for κ_+ and κ_- , it follows that

$$\frac{1}{\kappa_+} - \frac{1}{\kappa_-} = \frac{2 \frac{w}{c} A \sin(\omega t') \cos \theta}{1 - \left(\frac{w}{c}\right)^2 A^2 \sin^2(\omega t') \cos^2 \theta}, \quad (7.23)$$

$$\frac{1}{\kappa_+} + \frac{1}{\kappa_-} = \frac{2}{1 - \left(\frac{w}{c}\right)^2 A^2 \sin^2(\omega t') \cos^2 \theta}. \quad (7.24)$$

Now, we have reached the point where we will introduce our second approximation. Let's assume that charges which constitute our dipole move with non-relativistic velocities. Then $A\omega \ll c$, and the whole denominator in equations (7.23) and (7.24) becomes 1. Thus for "non-relativistic" electric dipole, the static (7.17) and the vector (7.18) potentials can be written as

$$\phi(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0 r} \left[2 \frac{w}{c} A \sin(\omega t') \cos \theta - 2 \left(\frac{A \cos(\omega t')}{r} \right)^1 P_1(\cos \theta) \right], \quad (7.25)$$

$$\mathbf{A}(\mathbf{r}, t) = -\frac{q\mathbf{v}(t')}{4\pi\epsilon_0 c^2 r} \left[2 - 2 \frac{w}{c} A \sin(\omega t') \cos \theta \left(\frac{A \cos(\omega t')}{r} \right)^1 P_1(\cos \theta) \right]. \quad (7.26)$$

Noting that $P_1(\cos \theta) = \cos \theta$, and $p_0 = q2A$, we finally write for scalar and vector potentials of an oscillating electric dipole:

$$\phi(\mathbf{r}, t) = \frac{p_0 \cos \theta}{4\pi\epsilon_0 r} \left[\frac{w}{c} \sin(\omega t') - \frac{\cos(\omega t')}{r} \right], \quad (7.27)$$

$$\mathbf{A}(\mathbf{r}, t) = \frac{p_0 \omega \sin(\omega t')}{4\pi\epsilon_0 c^2 r} \left[1 - \frac{w}{c} \frac{A^2 \sin(\omega t') \cos(\omega t') \cos^2 \theta}{r} \right] \mathbf{e}_x. \quad (7.28)$$

7.2 Electric and magnetic fields

Once we know the potentials, we can compute the electric and magnetic fields using already well known relations:

$$\mathbf{E} = -\nabla\phi - \frac{\partial\mathbf{A}}{\partial t}, \quad (7.29)$$

$$\mathbf{B} = \nabla \times \mathbf{A}. \quad (7.30)$$

Given the axial symmetry of this problem, a cylindrical coordinate system would be the natural choice for our calculations. However, since we have already introduced the polar angle θ from the spherical coordinate system (with the x-axis as the polar axis), we will continue using spherical coordinates for consistency in our subsequent derivations. In spherical coordinate system the nabla operator is defined as follows:

$$\nabla = \frac{\partial}{\partial r} \mathbf{e}_r + \frac{1}{r \sin \theta} \frac{\partial}{\partial \alpha} \mathbf{e}_\alpha + \frac{1}{r} \frac{\partial}{\partial \theta} \mathbf{e}_\theta. \quad (7.31)$$

Please note, that we have already used symbol ϕ for the scalar potential, so we had to choose something different to denote the azimuthal angle in spherical coordinates. α seemed to be a good choice.

Moreover, in our further considerations, we will only stick with the $1/r$ terms in the scalar and vector potentials. This is because our interest lies in the fields that persist at large distances from the dipole, and only these contributions lead to a Poynting vector that decays as $1/r^2$.

Thus, computing the radial component, and remembering that $t' = t - r/c$, we obtain:

$$\begin{aligned}\frac{\partial \phi}{\partial r} &= \frac{\partial}{\partial r} \left\{ \frac{p_0 \cos \theta}{4\pi\epsilon_0 r} \frac{w}{c} \sin(\omega t') \right\} \\ &= \frac{p_0 \cos \theta}{4\pi\epsilon_0} \left[-\frac{1}{r} \left(\frac{\omega}{c} \right)^2 \cos(\omega t') - \frac{1}{r^2} \left(\frac{\omega}{c} \right) \sin(\omega t') \right].\end{aligned}\quad (7.32)$$

Computing the azimuthal component:

$$\frac{1}{r \sin \theta} \frac{\partial \phi}{\partial \alpha} = \frac{1}{r \sin \theta} \frac{\partial}{\partial \alpha} \left\{ \frac{p_0 \cos \theta}{4\pi\epsilon_0 r} \frac{w}{c} \sin(\omega t') \right\} = 0. \quad (7.33)$$

Computing the polar component:

$$\frac{1}{r} \frac{\partial \phi}{\partial \theta} = \frac{1}{r} \frac{\partial}{\partial \theta} \left\{ \frac{p_0 \cos \theta}{4\pi\epsilon_0 r} \frac{w}{c} \sin(\omega t') \right\} = -\frac{p_0 \sin \theta}{4\pi\epsilon_0} \frac{1}{r^2} \left(\frac{w}{c} \right) \sin(\omega t'). \quad (7.34)$$

Therefore:

$$\begin{aligned}\nabla \phi &= \frac{\partial \phi}{\partial r} \mathbf{e}_r + \frac{1}{r \sin \theta} \frac{\partial \phi}{\partial \alpha} \mathbf{e}_\alpha + \frac{1}{r} \frac{\partial \phi}{\partial \theta} \mathbf{e}_\theta \\ &= -\frac{p_0}{4\pi\epsilon_0} \left[\frac{1}{r} \left(\frac{\omega}{c} \right)^2 \cos(\omega t') \cos \theta + \frac{1}{r^2} \left(\frac{\omega}{c} \right) \sin(\omega t') \cos \theta \right] \mathbf{e}_r \\ &\quad - \frac{p_0}{4\pi\epsilon_0} \left[\frac{1}{r^2} \left(\frac{\omega}{c} \right) \sin(\omega t') \sin \theta \right] \mathbf{e}_\theta.\end{aligned}\quad (7.35)$$

Likewise,

$$\begin{aligned}\frac{\partial \mathbf{A}}{\partial t} &= \frac{\partial}{\partial t} \left\{ \frac{p_0 \omega \sin(\omega t')}{4\pi\epsilon_0 c^2 r} \mathbf{e}_x \right\} = \frac{p_0 \omega^2 \cos(\omega t')}{4\pi\epsilon_0 c^2 r} (\cos \theta \mathbf{e}_r - \sin \theta \mathbf{e}_\theta) \\ &= \frac{p_0}{4\pi\epsilon_0} \left[\frac{1}{r} \left(\frac{\omega}{c} \right)^2 \cos(\omega t') \cos \theta \right] \mathbf{e}_r - \frac{p_0}{4\pi\epsilon_0} \left[\frac{1}{r} \left(\frac{\omega}{c} \right)^2 \cos(\omega t') \sin \theta \right] \mathbf{e}_\theta\end{aligned}\quad (7.36)$$

Thus

$$\begin{aligned}\mathbf{E} &= -\nabla \phi - \frac{\partial \mathbf{A}}{\partial t} = \frac{p_0}{4\pi\epsilon_0} \left[\frac{1}{r^2} \left(\frac{\omega}{c} \right) \sin(\omega t') \cos \theta \right] \mathbf{e}_r \\ &\quad + \frac{p_0}{4\pi\epsilon_0} \left[\frac{1}{r} \left(\frac{\omega}{c} \right)^2 \cos(\omega t') \sin \theta + \frac{1}{r^2} \left(\frac{\omega}{c} \right) \sin(\omega t') \sin \theta \right] \mathbf{e}_\theta.\end{aligned}\quad (7.37)$$

To calculate $\mathbf{B}$ we need to calculate $\nabla \times \mathbf{A}$. In spherical coordinate system, curl is defined as follows:

$$\begin{aligned} \nabla \times \mathbf{A} = & \frac{1}{r \sin \theta} \left(\frac{\partial (\sin \theta A_\alpha)}{\partial \theta} - \frac{\partial A_\theta}{\partial \alpha} \right) \mathbf{e}_r \\ & + \frac{1}{r} \left(\frac{\partial (r A_\theta)}{\partial r} - \frac{\partial A_r}{\partial \theta} \right) \mathbf{e}_\alpha + \frac{1}{r} \left(\frac{1}{\sin \theta} \frac{\partial A_r}{\partial \alpha} - \frac{\partial (r A_\alpha)}{\partial r} \right) \mathbf{e}_\theta \end{aligned} \quad (7.38)$$

Because $A_\alpha = 0$ and $\frac{\partial A_\theta}{\partial \alpha} = \frac{\partial A_r}{\partial \alpha} = 0$, we are only left with the azimuthal term for which we get:

$$\begin{aligned} & \frac{1}{r} \left(\frac{\partial (r A_\theta)}{\partial r} - \frac{\partial A_r}{\partial \theta} \right) \mathbf{e}_\alpha \\ &= \frac{1}{r} \left(\frac{p_0 \omega^2}{4\pi \epsilon_0 c^3} \cos(\omega t') \sin \theta + \frac{p_0 \omega}{4\pi \epsilon_0 c^2} \frac{1}{r} \sin(\omega t') \sin \theta \right) \mathbf{e}_\alpha \\ &= \frac{p_0}{4\pi \epsilon_0} \left(\frac{1}{r} \frac{\omega^2}{c^3} \cos(\omega t') \sin \theta + \frac{1}{r^2} \frac{\omega}{c^2} \sin(\omega t') \sin \theta \right) \mathbf{e}_\alpha \end{aligned} \quad (7.39)$$

To sum up this chapter, let me present once again in one table the electric and magnetic fields of an oscillating electric dipole. These are:

$$\begin{aligned} \mathbf{E} = & \frac{p_0}{4\pi \epsilon_0} \left[\frac{1}{r^2} \left(\frac{\omega}{c} \right) \sin(\omega t') \cos \theta \right] \mathbf{e}_r \\ & + \frac{p_0}{4\pi \epsilon_0} \left[\frac{1}{r} \left(\frac{\omega}{c} \right)^2 \cos(\omega t') \sin \theta + \frac{1}{r^2} \left(\frac{\omega}{c} \right) \sin(\omega t') \sin \theta \right] \mathbf{e}_\theta \end{aligned} \quad (7.40)$$

$$\mathbf{B} = \frac{p_0}{4\pi \epsilon_0} \left[\frac{1}{r} \frac{\omega^2}{c^3} \cos(\omega t') \sin \theta + \frac{1}{r^2} \frac{\omega}{c^2} \sin(\omega t') \sin \theta \right] \mathbf{e}_\alpha \quad (7.41)$$

7.3 The Poynting vector

To calculate the radiated power from an oscillating electric dipole, we construct the Poynting vector using only the $1/r$ terms from the electric and magnetic fields:

$$\mathbf{E} = \frac{p_0}{4\pi \epsilon_0} \left[\frac{1}{r} \left(\frac{\omega}{c} \right)^2 \cos(\omega t') \sin \theta \right] \mathbf{e}_\theta, \quad (7.42)$$

$$\mathbf{B} = \frac{p_0}{4\pi \epsilon_0} \left[\frac{1}{r} \frac{\omega^2}{c^3} \cos(\omega t') \sin \theta \right] \mathbf{e}_\alpha. \quad (7.43)$$

Note that such "reduced" $\mathbf{E}$ and $\mathbf{B}$ fields are in phase, mutually perpendicular to each other, and their magnitudes satisfy the relation $\frac{|\mathbf{E}|}{|\mathbf{B}|} = c$.

The Poynting vector is then

$$\begin{aligned}\mathbf{S}(\mathbf{r}, t) &= \mathbf{E}(\mathbf{r}, t) \times \mathbf{H}(\mathbf{r}, t) \\ &= c^2 \epsilon_0 \mathbf{E}(\mathbf{r}, t) \times \mathbf{B}(\mathbf{r}, t) = \frac{p_0^2 \omega^4}{16\pi^2 \epsilon_0 c^3} \left[\frac{1}{r} \cos(\omega t') \sin \theta \right]^2 \mathbf{e}_r.\end{aligned}\quad (7.44)$$

The average value of the cosine is $1/2$. Therefore, the time-averaged (over one cycle) Poynting vector is

$$\langle S(\mathbf{r}) \rangle = \frac{p_0^2 \omega^4}{32\pi^2 \epsilon_0 c^3} \left(\frac{1}{r^2} \sin^2 \theta \right) \mathbf{e}_r. \quad (7.45)$$

7.4 Radiated power

The total radiated power through a sphere of radius r is obtained by integrating the time-averaged Poynting vector (7.45) over all solid angles

$$\begin{aligned}\langle P \rangle &= \int \langle S(\mathbf{r}) \rangle \cdot d\mathbf{A} = \int r^2 \langle S(\mathbf{r}) \rangle \cdot d\Omega \\ &= \frac{p_0^2 \omega^4}{32\pi^2 \epsilon_0 c^3} \int_0^\pi \sin^3 \Theta d\Theta \int_0^{2\pi} d\phi = \frac{p_0^2 \omega^4}{32\pi^2 \epsilon_0 c^3} \cdot \frac{4}{3} \cdot 2\pi = \frac{p_0^2 \omega^4}{12\pi \epsilon_0 c^3}.\end{aligned}\quad (7.46)$$

If we now recall the Larmor formula, which gives the total radiation power for non-relativistic particles

$$P(t) = \frac{q^2 a^2(t')}{6\pi \epsilon_0 c^3}, \quad (7.47)$$

and notice that $a(t') = \pm A\omega^2 \cos(\omega t')$ for each of the particles constituting a dipole, then $\langle a \rangle = \frac{1}{2} A\omega^2$, and

$$\langle P_+ \rangle = \langle P_- \rangle = \frac{q^2 \langle a \rangle^2}{6\pi \epsilon_0 c^3} = \frac{q^2 A^2 \omega^4}{24\pi \epsilon_0 c^3}. \quad (7.48)$$

Finally, the total radiated power from an oscillating dipole, obtained from the Larmor formula, is

$$\langle P \rangle = \langle P_+ \rangle + \langle P_- \rangle = \frac{q^2 A^2 \omega^4}{12\pi \epsilon_0 c^3} = \frac{p_0^2 \omega^4}{12\pi \epsilon_0 c^3}, \quad (7.49)$$

which aligns perfectly with the result obtained by direct integration of the Poynting vector over a sphere of radius r (equation (7.46)).

7.5 Visualizations of $\mathbf{E}$, $\mathbf{B}$ and $\mathbf{S}$ fields

7.5.1 Point charge moving with non-relativistic velocities

The visualizations presented below illustrate the electric field $\mathbf{E}$, magnetic field $\mathbf{B}$, and the Poynting vector $\mathbf{S}$ of an oscillating electric dipole operating at a frequency of $f = 1.20 \times 10^8$ Hz, which corresponds to a wavelength of approximately $\lambda \approx 2.5$ m. At this frequency, the dipole oscillates with non-relativistic velocities, meaning $A\omega \ll c$.

The electric field exhibits the characteristic dipole pattern with field lines originating from the positive charge and terminating on the negative charge, whereas the Poynting vector exhibits the characteristic "dumbbell" shape typical of dipole radiation. Furthermore, there is no radiation along the axis of the dipole ($\theta = 0$) as expected from the analytical solutions. See equation (7.44).

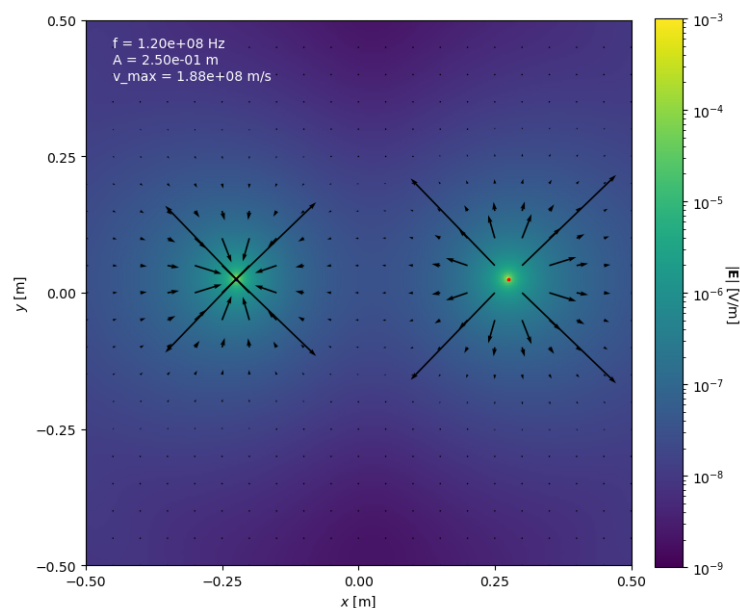


Figure 7.2: Electric field $\mathbf{E}$ of an oscillating electric dipole moving with non-relativistic velocity.

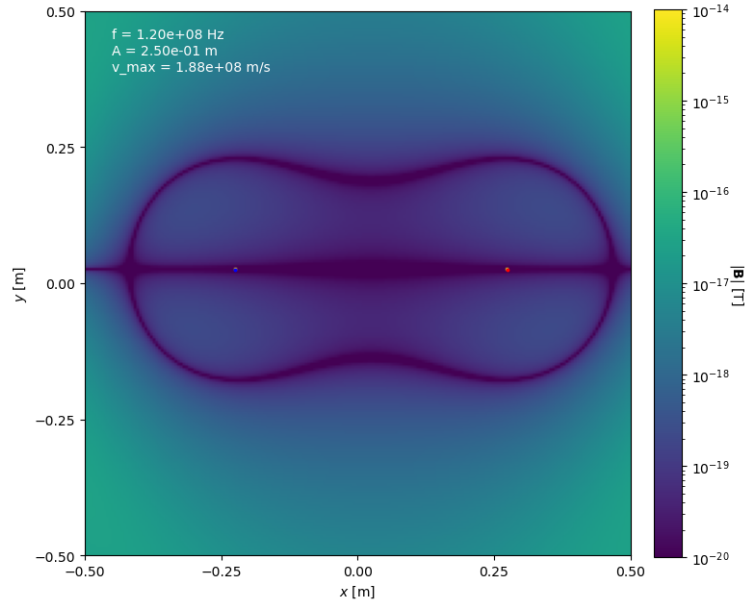


Figure 7.3: Magnetic field $\mathbf{B}$ of an oscillating electric dipole moving with non-relativistic velocity.

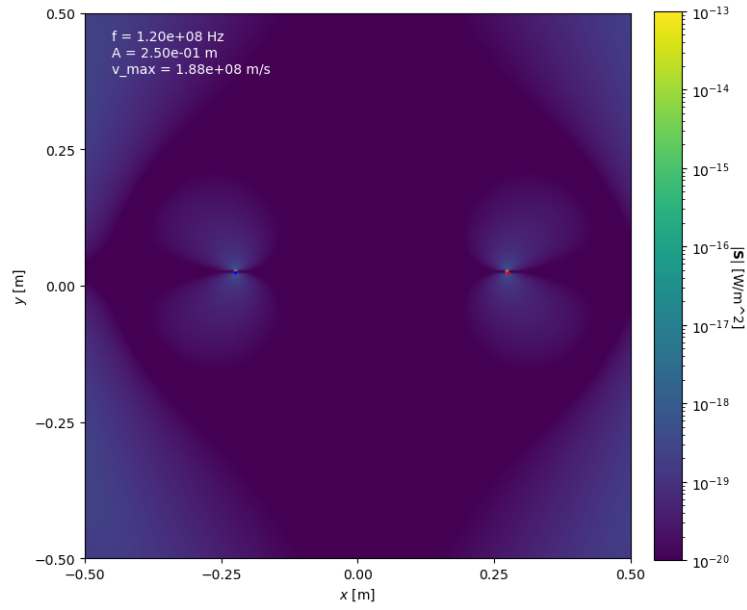


Figure 7.4: Poynting vector $\mathbf{S}$ of an oscillating electric dipole moving with non-relativistic velocity.

7.5.2 Point charge moving with relativistic velocities

The visualizations presented below illustrate the electric field $\mathbf{E}$, magnetic field $\mathbf{B}$, and the Poynting vector $\mathbf{S}$ of an oscillating electric dipole operating at a frequency of $f = 9.00 \times 10^8$ Hz, which corresponds to a wavelength of approximately $\lambda \approx 0.33$ m. At this frequency, the dipole oscillates with relativistic velocities, meaning $A\omega \sim c$. But, if you carefully look at the attached plots, you will notice that the wavelength is not ≈ 0.33 m. It is half of that. So why is that? This is because in our derivations in this chapter we have introduced few approximations which are valid for far zones, and more importantly, for particles moving with non-relativistic velocities. At the early stages of derivations, when dealing with scalar and vector potentials, we've got rid of terms like $\left(\frac{w}{c}\right)^2 A^2 \sin^2(\omega t') \cos^2 \theta$, which actually are predominant for relativistic velocities. And these are the terms which introduce this doubled frequency visible on the attached figures.

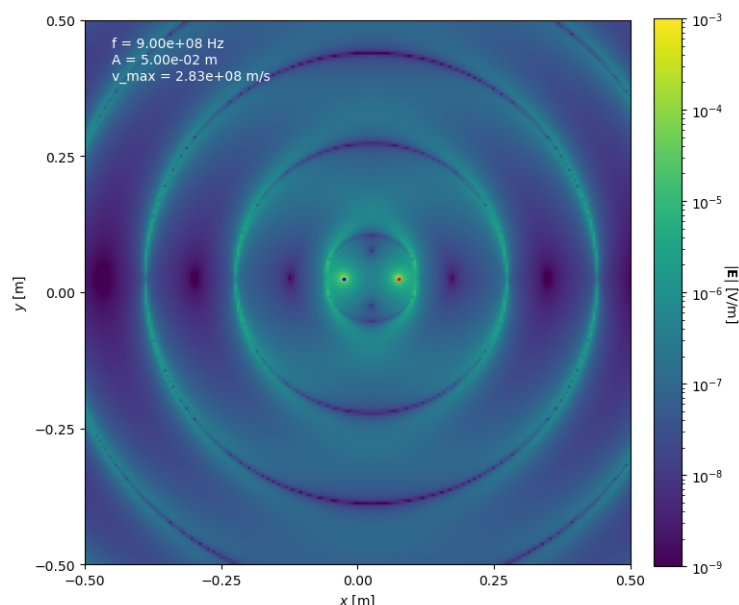


Figure 7.5: Electric field $\mathbf{E}$ of an oscillating electric dipole moving with relativistic velocity.

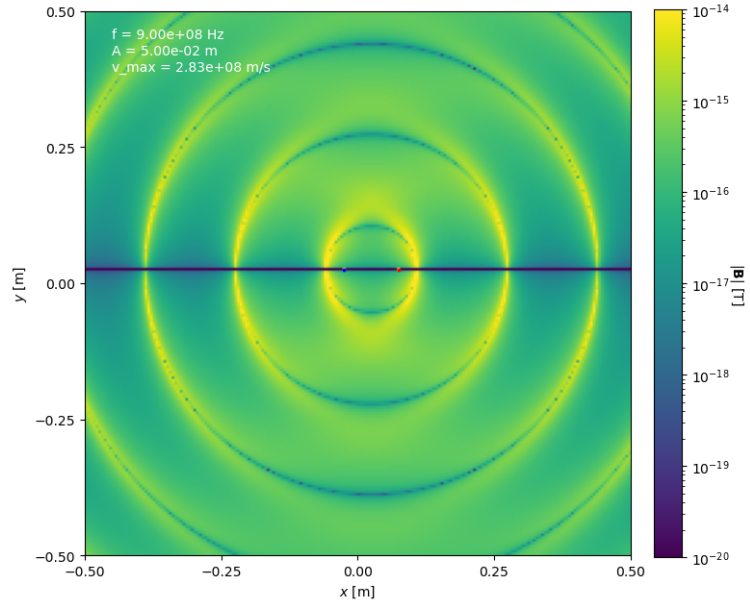


Figure 7.6: Magnetic field $\mathbf{B}$ of an oscillating electric dipole moving with relativistic velocity.

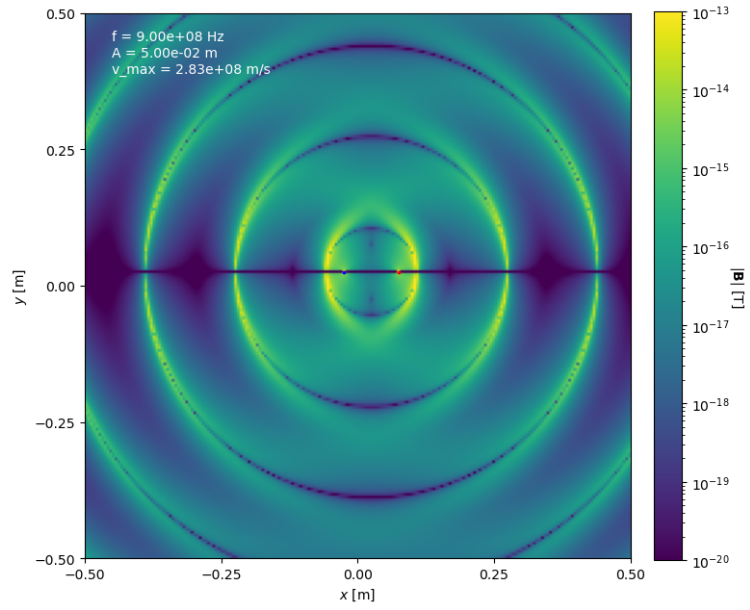


Figure 7.7: Poynting vector $\mathbf{S}$ of an oscillating electric dipole moving with relativistic velocity.

8 Radiation from a point charge moving with constant angular velocity

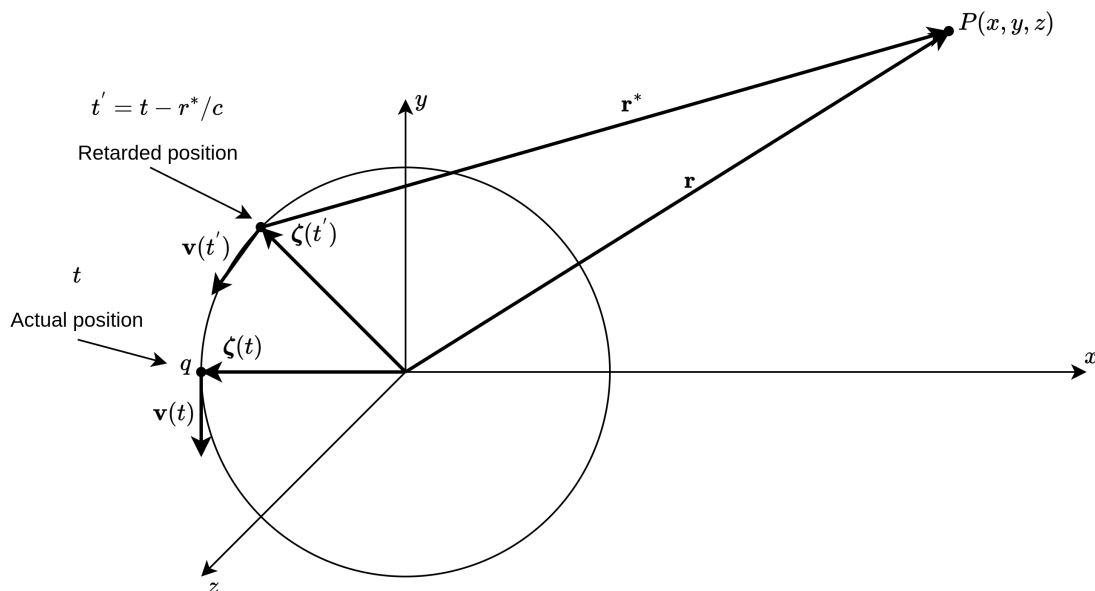


Figure 8.1: Point charge moving with constant angular velocity.

8.1 Radiated power

We will start this discussion by yet again recalling equation (4.67) from chapter 4:

$$P(t') = -\gamma^6 \frac{q^2}{6\pi\epsilon_0 c^3} \left[a^2(t') - \frac{1}{c^2} \left(\mathbf{v}(t') \times \mathbf{a}(t') \right)^2 \right], \quad (8.1)$$

where $P(t') = \frac{d\mathcal{E}}{dt'}$ is the instantaneous power radiated by a charged particle moving with a relativistic velocity.

In the case of uniform circular motion $\mathbf{v}(t')$ is all the time perpendicular to $\mathbf{a}(t')$, thus $\mathbf{v}(t') \times \mathbf{a}(t') = v(t')a(t')$ and the equation (4.67) takes the following form:

$$P(t') = -\gamma^4 \frac{q^2}{6\pi\epsilon_0 c^3} a^2(t'). \quad (8.2)$$

Let us now consider the case where the point charge q with the rest mass m_0 moves in the circular orbit caused by the centripetal electrostatic force of a heavy nucleus with the charge Zq positioned in the center of the orbit. Then

$$a(t') = \frac{F}{\gamma m_0} = \frac{Zq^2}{4\pi\epsilon_0} \frac{1}{r^2 \gamma m_0}, \quad (8.3)$$

where r is the radius of that orbit.

Substituting (8.3) into (8.2) yields the following expression for radiated power

$$P(t') = -\gamma^2 \frac{q^2}{6\pi\epsilon_0 c^3} \frac{Z^2 q^4}{(4\pi\epsilon_0)^2} \frac{1}{r^4 m_0^2}. \quad (8.4)$$

Expression (8.4) depends on two variables: the charge's velocity (hidden in the γ factor) and the charge's radius. Let's try to make it dependent only on one variable and let that variable be the charge's energy itself (or any function of the energy). That way we shall receive some form of differential equation with the charge's energy as the unknown/searched function. To do so, we will start by recalling the equation for the total energy of a charge moving in the field of a (much heavier) nucleus. And because we are considering a relativistic case, we shall also use a relativistic version of such an equation. For the relativistic case we have to add the $m_0 c^2$ term to the usual sum of the charge's kinetic and potential energies. This is

$$\mathcal{E} = m_0 c^2 + \mathcal{E}_k + \mathcal{E}_p = m_0 c^2 + (\gamma - 1) m_0 c^2 - \frac{Zq^2}{4\pi\epsilon_0 r} = \gamma m_0 c^2 - \frac{Zq^2}{4\pi\epsilon_0 r}. \quad (8.5)$$

Please note that we have also used the relativistic version for the kinetic energy $(\gamma - 1) m_0 c^2$ instead of the classical $\frac{mv^2}{2}$ as this is exactly the case we are considering (relativistic particle moving in the electrostatic field of a heavy nucleus).

The second equation, which will be very helpful in our derivations, comes from Newton's 2nd law (we have used it already in (8.3)):

$$\mathbf{F} = \gamma m_0 \mathbf{a} \rightarrow \frac{Zq^2}{4\pi\epsilon_0 r^2} = \gamma m_0 \frac{v^2}{r}. \quad (8.6)$$

The electrostatic force between nucleus and the orbiting charge is constrained to equal the centripetal force required to maintain such motion.

Armed with equations (8.5) and (8.6) we write for the charge total energy:

$$\mathcal{E} = \gamma m_0 c^2 - \gamma m_0 v^2 = \frac{1}{\gamma} m_0 c^2. \quad (8.7)$$

Thus

$$\gamma = \frac{m_0 c^2}{\mathcal{E}}. \quad (8.8)$$

If we now substitute (8.8) into the expression for the kinetic energy $\mathcal{E}_k$ we will get

$$\mathcal{E}_k = (\gamma - 1) m_0 c^2 = \left(\frac{m_0 c^2}{\mathcal{E}} - 1 \right) m_0 c^2 \quad (8.9)$$

and the expression (8.5) can be written as

$$\mathcal{E} = \frac{(m_0 c^2)^2}{\mathcal{E}} - \frac{Z q^2}{4\pi\epsilon_0 r}. \quad (8.10)$$

Thus

$$\frac{Z q^2}{4\pi\epsilon_0 r} = \frac{(m_0 c^2)^2}{\mathcal{E}} - \mathcal{E}, \quad (8.11)$$

and

$$r = \frac{Z q^2}{4\pi\epsilon_0} \frac{\mathcal{E}}{(m_0 c^2)^2 - \mathcal{E}^2}. \quad (8.12)$$

Let us digress a bit at that moment and do some verification. We know that the rest energy of the electron is $m_e c^2$. The energy of the electron in the ground (bound) state in the hydrogen atom is reduced by the binding energy, which in this case is -13.6 eV. Having obtained the energy, let us calculate the radius r using the formula (8.12). For $\mathcal{E} = m_e c^2 - 13.6$ eV ≈ 510985.350692 eV we get $r \approx 0.05293917$ nm which corresponds very well to the radius of the electron in the ground state in the Bohr model of the hydrogen atom. Let's come back to derivations.

Substituting (8.8) and (8.12) into (8.4) we finally get

$$\frac{d\mathcal{E}}{dt'} = - \frac{(m_0 c^2)^2}{\mathcal{E}^2} \frac{q^2}{6\pi\epsilon_0 c^3} \frac{Z^2 q^4}{(4\pi\epsilon_0)^2} \frac{\left((m_0 c^2)^2 - \mathcal{E}^2 \right)^4 (4\pi\epsilon_0)^4}{\mathcal{E}^4 Z^4 q^8 m_0^2}, \quad (8.13)$$

which after reductions takes the following form

$$\frac{d\mathcal{E}}{dt'} = - \frac{8}{3} \frac{c\pi\epsilon_0}{Z^2 q^2} \frac{\left((m_0 c^2)^2 - \mathcal{E}^2 \right)^4}{\mathcal{E}^6}. \quad (8.14)$$

This is our differential equation describing energy loss by an orbiting (and thus radiating) point charge in the electrostatic field of a heavy nucleus.

Let's assume now that we are dealing with the classical "Bohr" hydrogen atom ($Z = 1$). Using SI values

- $c = 2.99792458 \times 10^8$ m/s
- $\epsilon_0 = 8.8541878128 \times 10^{-12}$ F/m
- $q = 1.602176634 \times 10^{-19}$ C

we get

$$\frac{8}{3} \frac{c\pi\epsilon_0}{Z^2 q^2} = 8.659 \times 10^{35} \frac{1}{\text{J} \cdot \text{s}} = 1.387 \times 10^{17} \frac{1}{\text{eV} \cdot \text{s}}. \quad (8.15)$$

If the electron's starting energy is $m_e c^2 - 13.6$ eV (which is the electron energy in the ground state), then

$$\frac{\left((m_e c^2)^2 - \mathcal{E}^2\right)^4}{\mathcal{E}^6} = 2.096 \times 10^{-6} (\text{eV})^2 \quad (8.16)$$

and the whole energy loss is

$$\frac{d\mathcal{E}}{dt'} = 2.907 \times 10^{11} \frac{\text{eV}}{\text{s}}. \quad (8.17)$$

This is a tremendous value, and it is getting even bigger as electron approaches the nucleus. With that rate of loss of energy, the electron would spiral into the nucleus in a few picoseconds. So, where is the problem? Is the whole electrodynamics wrong? After all we know that hydrogen atom is stable (it does not collapse). Even more, the energy (if emitted), is not emitted continuously. From experiments we know that atoms emit only discrete spectral lines (Balmer series, Lyman series, etc.). The short answer to the above questions is: **At the atomic level the classical electrodynamics fails miserably.** That problem of continuous radiation when accelerating, together with "ultraviolet catastrophe" and discovery of photoelectric effect eventually led to formulation of quantum mechanics. But that's another tale.

8.2 Visualizations of E, B and S fields

8.2.1 Point charge moving with non-relativistic velocities

Three figures below show the electric field, magnetic field and Poynting vector for a point charge orbiting with constant frequency of 1.00×10^8 Hz around a center, with the radius of $r = 4.50 \times 10^{-2}$ m. This gives constant angular velocity $\omega = 2\pi f = 6.26 \times 10^8$ rad/s, and linear velocity $v = \omega r = 2.83 \times 10^7$ m/s. This is

around 9 % of the speed of light. With that velocity, the fields do not differ much from the fields generated by a stationary charge, apart from the fact that their spatial distribution spans now from the instantaneous position of the charge, which is orbiting around a center. If interested, please have a look at animations available at <https://github.com/lukasz-wieczaszek/radiation-by-moving-charges> in the folder `charge-moving-with-constant-angular-velocity`.

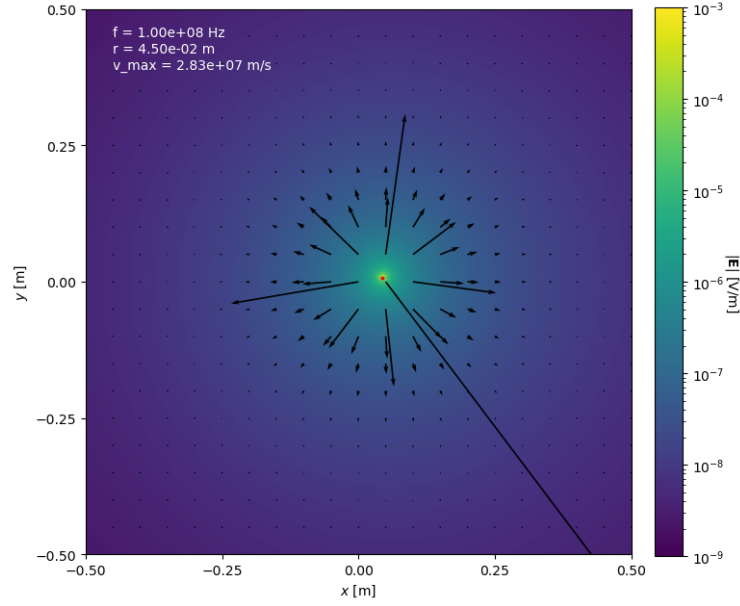


Figure 8.2: Electric field $\mathbf{E}$ of a point charge moving with constant non-relativistic angular velocity.

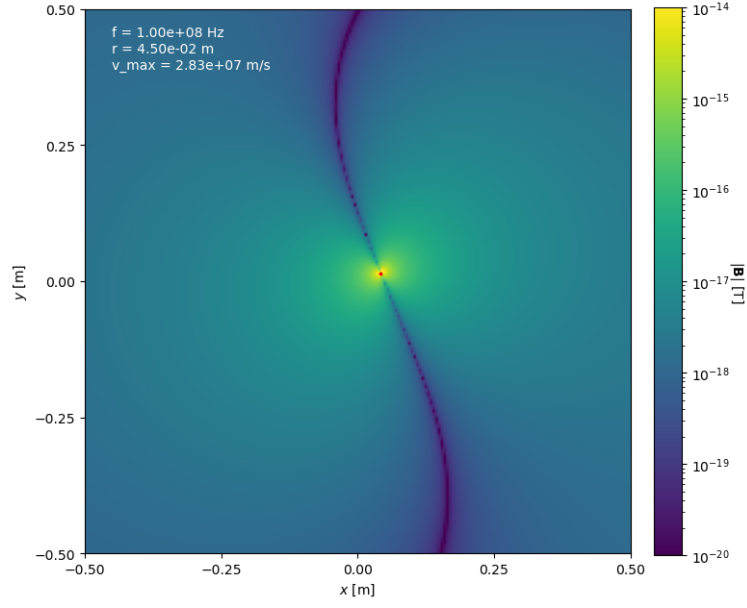


Figure 8.3: Magnetic field $\mathbf{B}$ of a point charge moving with constant non-relativistic angular velocity.

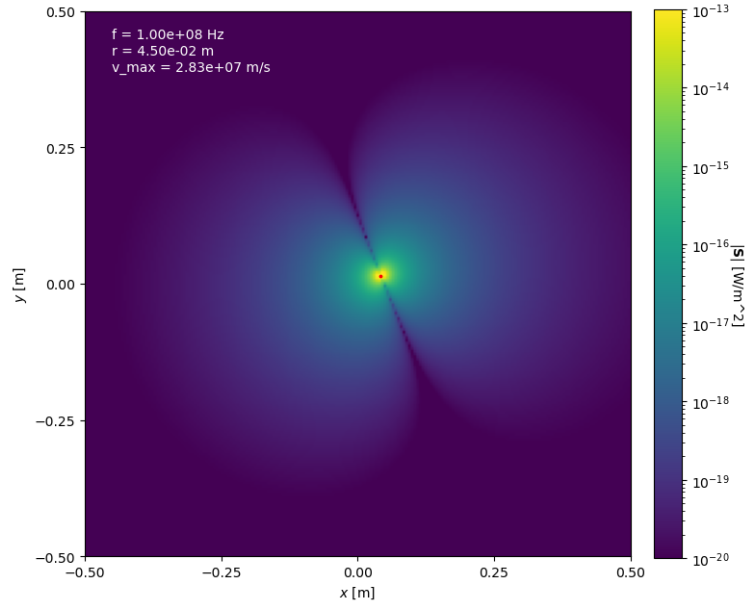


Figure 8.4: Poynting vector $\mathbf{S}$ of a point charge moving with constant non-relativistic angular velocity.

8.2.2 Point charge moving with relativistic velocities

The situation changes drastically if we increase the angular velocity so that linear velocity is now close to the speed of light. With the frequency of 9.00×10^8 Hz the angular velocity $\omega = 2\pi f = 5.65 \times 10^9$ rad/s, and linear velocity $v = \omega r = 2.54 \times 10^8$ m/s. The radius r stays the same ($r = 4.50 \times 10^{-2}$ m). Now the charge is moving with the speed, which is 85 % of the speed of light. With that speed, the electric and magnetic fields as well as the Poynting vector reveal a spiral pattern which propagates outward from the center around which the charge revolves.

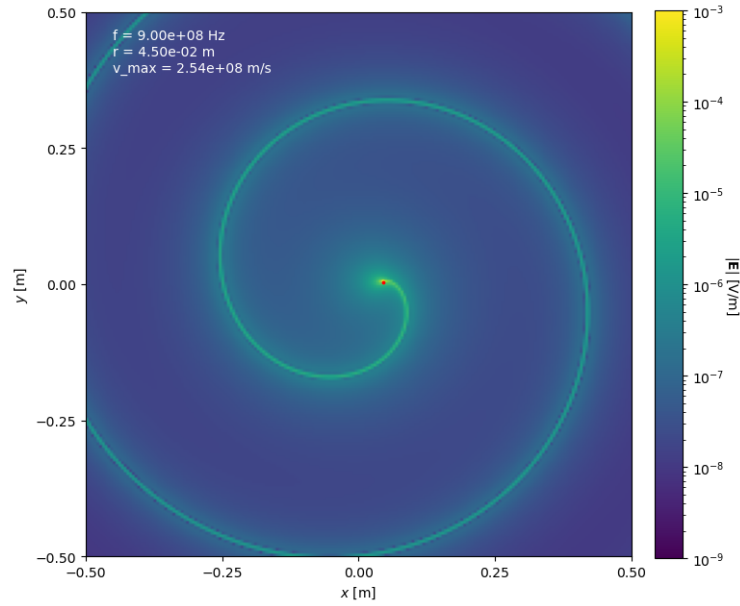


Figure 8.5: Electric field $\mathbf{E}$ of a point charge moving with constant relativistic angular velocity.

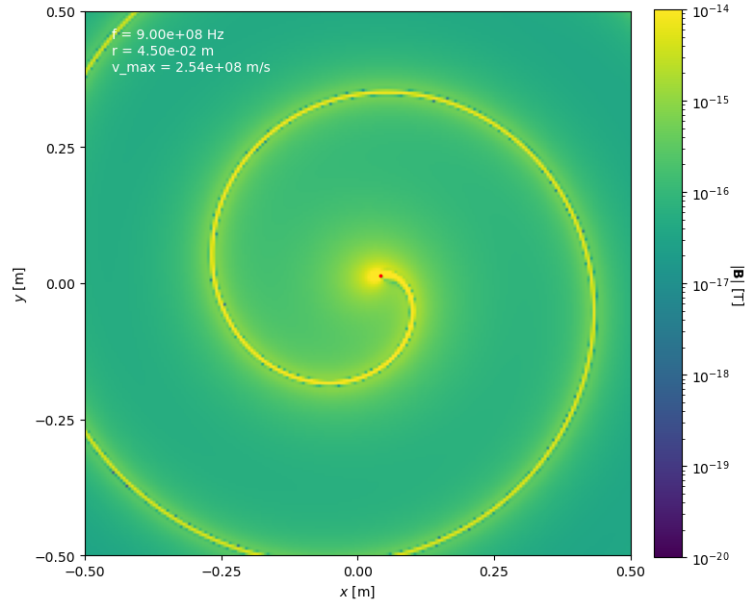


Figure 8.6: Magnetic field $\mathbf{B}$ of a point charge moving with constant relativistic angular velocity.

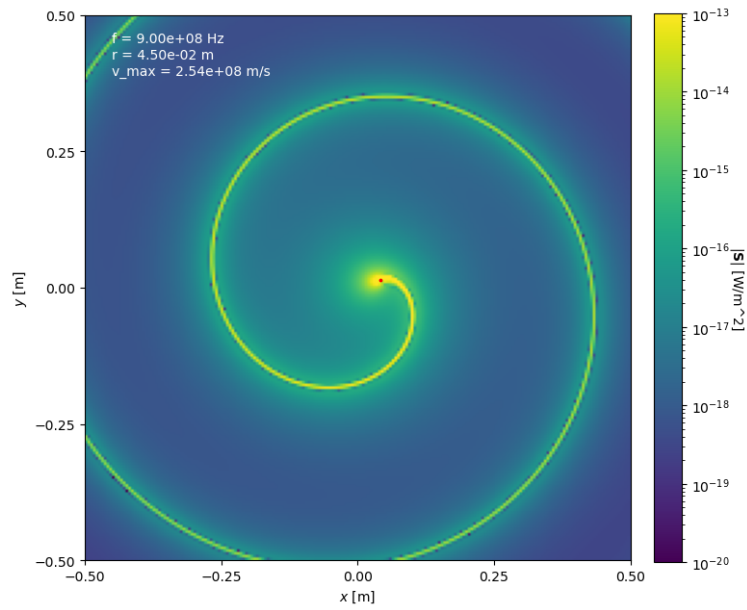


Figure 8.7: Poynting vector $\mathbf{S}$ of a point charge moving with constant relativistic angular velocity.

9 Free falling

While drafting "Oh, well" in one of my papers, Peter Green instantly sprang to mind. It looks that scenario has repeated itself, though this time it's Mr. Tom Petty who has triggered that persistent earworm.

Nevertheless, let's try to return to the fundamental question. Does a freely falling charged particle radiate electromagnetic radiation? After all, if it falls freely, it has acceleration, and if it has acceleration, then according to the Larmor formula (4.63, 4.67), it loses the energy that is emitted and propagated via an electromagnetic wave. And this is not only theoretical work. There are plenty of examples where this occurs. Unfortunately, all of them that are known to me are caused by non-gravitational forces. Starting from the most obvious ones surrounding us on all sides — radio waves, which are created by accelerating electrons in transmitting antennas, through synchrotron facilities and x-ray tubes, to more sophisticated Thomson scattering experiments or Free-Electron Lasers (FELs). So one could just terminate further considerations and say that we have so many examples and experiments that prove the validity of Larmor formula that adopting it to acceleration due to gravitation does not change much. If it is accelerating, then it is radiating. That's all. But let me show how many problems and questions such an approach brings.

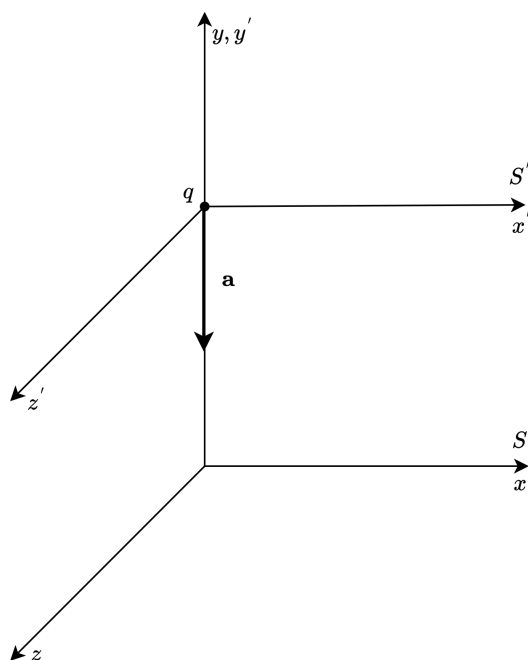


Figure 9.1: Point charge freely falling in S but fixed in S' .

Let's consider a freely falling (relative to some "laboratory" frame S) reference frame S' . In the general theory of relativity, a freely falling frame is an inertial frame. Now imagine a charged particle at rest that is bound to the origin of such a frame. For any observer inside S' , the particle is just standing still and produces an ordinary "Coulomb" electric field. Nothing more. No magnetic field, no radiation. But for an observer in the "laboratory" frame S , the particle is moving/falling with constant acceleration, and, according to the Larmor formula, emits electromagnetic radiation (if we blindly extend all considerations from classical electrodynamics). But this raises a sequence of additional questions:

- Does the principle of equivalence break down here? After all, the effects produced by a charged particle freely falling in a gravitational field should be indistinguishable from those of the same charge at rest in empty space. And we have already established that a stationary charge does not emit radiation.
- If the particle radiates, what is the source of the energy carried by the electromagnetic wave?
- If an electromagnetic wave induces effects in frame S , let's say it generates a current in a receiving antenna, how would this induction be interpreted in frame S' ? What phenomena would account for this?
- And most importantly, are there any experiments showing that freely falling charged particle in fact radiates energy?

It appears that this paradox remains subtle because isolating a purely gravitational acceleration from other effects is extremely challenging. Most practical radiation scenarios involve non-gravitational forces that clearly produce observable effects. I think that a good explanation to the problem is provided in [3]. The acceleration that matters and is in fact the reason of the electromagnetic radiation is the relative acceleration between the charge and its own electric field. That is why a freely falling charge does not produce an EM radiation. A freely falling charge in a gravitational field follows geodesic lines the same way as the electric field does. That way, there is no relative motion/acceleration between the charge and its electric field the same way as there is no relative motion between a static/fixed charge and its field in the "empty" space, and the principle of equivalence is not violated. But what is the source of the energy when the particle does radiate? It turns out that this can be explained by the curvature of electric field lines which appears when a charge accelerates relative to its own electric field. This curvature gives rise to a stress force, and the work done to overcome the stress force is the source of the energy carried by the radiation. Please refer also to [2].

10 Mathematical supplement

10.1 Solution of the inhomogeneous wave equation

Equation

$$\left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}\right) \xi(\mathbf{r}, t) = -f(\mathbf{r}, t). \quad (10.1)$$

where $\xi(\mathbf{r}, t)$ is an unknown "potential" function, and $f(\mathbf{r}, t)$ is some known so-called "forcing/source" function is an inhomogeneous three-dimensional wave equation. Let us try to find its solution. We will start this process by recalling the Laplace transform $\mathcal{L}$.

The Laplace transform $\mathcal{L}$ of a function $f(t)$, defined for all real numbers $t \geq 0$, is the function $F(s)$, which is given by the following formula

$$F(s) = \int_0^\infty f(t) e^{-st} dt, \quad (10.2)$$

where s is a complex number, often represented in the form $s = \sigma + i\omega$, with σ and ω being real numbers. Thus

$$F(s) = \mathcal{L}\{f(t)\}$$

and

$$f(t) = \mathcal{L}^{-1}\{F(s)\}.$$

Using the above notation we write

$$\mathcal{L}\{f(\mathbf{r}, t)\} = F(\mathbf{r}, s)$$

and

$$\mathcal{L}\left\{\left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}\right) \xi(\mathbf{r}, t)\right\} = \left(\nabla^2 - \frac{s^2}{c^2}\right) \Xi(\mathbf{r}, s).$$

In the last transformation we used the property of the Laplace transform which says that differentiation in the time domain corresponds to a multiplication by s in the s -domain

$$\mathcal{L}\{f''(t)\} = s^2 F(s) - sf(0) - f'(0),$$

plus we used the assumption that $f(0) = 0$.

Our inhomogeneous wave equation in the time domain (10.1) becomes an inhomogeneous Helmholtz' equation in the s -domain.

$$(\nabla^2 + k^2) \Xi(\mathbf{r}, s) = -F(\mathbf{r}, s) \quad (10.3)$$

where $k = i\frac{s}{c}$, and i is an imaginary number.

Thus, let us try to find a Green's function $G(\mathbf{r}, \mathbf{r}')$ for the Helmholtz' operator $\nabla^2 + k^2$ and try to use Green's function method of solution.

To find a Green's function we need to solve the following equation

$$(\nabla^2 + k^2) G(\mathbf{r}, \mathbf{r}') = \delta^3(\mathbf{r} - \mathbf{r}'). \quad (10.4)$$

To solve that equation on the other hand, we need to impose some boundary conditions. And because Ξ represents electric potential, it is very plausible to adopt condition where $G(\mathbf{r}, \mathbf{r}') \rightarrow 0$ as $|\mathbf{r} - \mathbf{r}'| \rightarrow \infty$. That way potential goes to zero as distance from the source charge goes to infinity. We also notice that the equation (10.4) is spherically symmetric about the point $\mathbf{r}'$. That way Green's function shall also be spherically symmetric about the point $\mathbf{r}'$, so that we may write $G(\mathbf{r}, \mathbf{r}') = G(|\mathbf{r} - \mathbf{r}'|)$.

The Laplace operator in spherical coordinates is

$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \phi^2}. \quad (10.5)$$

If we introduce $\tilde{\mathbf{r}} = \mathbf{r} - \mathbf{r}'$, $\tilde{r} = |\tilde{\mathbf{r}}|$, and express equation (10.4) in the spherical coordinates (taking into account spherical symmetry about the point $\mathbf{r}'$) it will reduce to

$$\frac{1}{\tilde{r}^2} \frac{\partial}{\partial \tilde{r}} \left(\tilde{r}^2 \frac{\partial G(\tilde{r})}{\partial \tilde{r}} \right) + k^2 G(\tilde{r}) = \delta^3(\tilde{\mathbf{r}}) \quad (10.6)$$

The solution to the above equation is

$$G(\tilde{r}) = -\frac{1}{4\pi} \frac{e^{ik\tilde{r}}}{\tilde{r}}. \quad (10.7)$$

Let us verify that. For $\tilde{r} \neq 0$ we get

$$\begin{aligned} \frac{1}{\tilde{r}^2} \frac{\partial}{\partial \tilde{r}} \left(\tilde{r}^2 \frac{\partial}{\partial \tilde{r}} \left(-\frac{e^{ik\tilde{r}}}{4\pi\tilde{r}} \right) \right) - k^2 \frac{e^{ik\tilde{r}}}{4\pi\tilde{r}} &= \frac{1}{\tilde{r}^2} \frac{\partial}{\partial \tilde{r}} \left(\tilde{r}^2 \frac{(1 - ik\tilde{r}) e^{ik\tilde{r}}}{4\pi\tilde{r}^2} \right) - k^2 \frac{e^{ik\tilde{r}}}{4\pi\tilde{r}} \\ &= \frac{1}{\tilde{r}^2} \frac{k^2 \tilde{r} e^{ik\tilde{r}}}{4\pi} - k^2 \frac{e^{ik\tilde{r}}}{4\pi\tilde{r}} = 0 \end{aligned}$$

But for $\tilde{r} = 0$, the expression (10.7) is not defined. Therefore we have to perform a limiting procedure. Let's consider a small volume, bounded by surface S , of radius ϵ , centered on $\mathbf{r}'$. We integrate $(\nabla^2 + k^2) G(\tilde{r})$ in the neighborhood of $\mathbf{r}'$, and transform this volume integral by means of Gauss' theorem into a surface integral independent of $\tilde{r}$.

$$\iiint_V (\nabla^2 + k^2) G(\tilde{r}) dV = \iiint_V \nabla^2 G(\tilde{r}) dV + \iiint_V k^2 G(\tilde{r}) dV = I_1 + I_2$$

$$\begin{aligned} I_1 &= \iiint_V \nabla \cdot (\nabla G(\tilde{r})) dV = \oiint_S \nabla G(\tilde{r}) \cdot d\mathbf{S} = \oiint_S \left(\frac{\partial}{\partial \tilde{r}} \left(-\frac{e^{ik\tilde{r}}}{4\pi\tilde{r}} \right) \right) \tilde{r}^2 d\Omega \\ &= \oiint_S \frac{(1 - ik\tilde{r}) e^{ik\tilde{r}}}{4\pi\tilde{r}^2} \tilde{r}^2 d\Omega = \frac{(1 - ik\tilde{r}) e^{ik\tilde{r}}}{4\pi} \oiint_S d\Omega = (1 - ik\tilde{r}) e^{ik\tilde{r}} \end{aligned}$$

$$\begin{aligned} I_2 &= \int_0^\epsilon \oint_0^{4\pi} k^2 G(\tilde{r}) \tilde{r}^2 d\tilde{r} d\Omega = \int_0^\epsilon \oint_0^{4\pi} k^2 \left(-\frac{e^{ik\tilde{r}}}{4\pi\tilde{r}} \right) \tilde{r}^2 d\tilde{r} d\Omega \\ &= \int_0^\epsilon k^2 \left(-\frac{e^{ik\tilde{r}}}{4\pi\tilde{r}} \right) \tilde{r}^2 d\tilde{r} \oint_0^{4\pi} d\Omega = -k^2 \int_0^\epsilon \tilde{r} e^{ik\tilde{r}} d\tilde{r} \\ &= -k^2 \frac{(1 - ik\tilde{r}) e^{ik\tilde{r}}}{k^2} \bigg|_0^\epsilon = -1 + (1 - ik\epsilon) e^{ik\epsilon} \end{aligned}$$

As $\epsilon \rightarrow 0$, so does $\tilde{r}$. Then $I_1 \rightarrow 1$, $I_2 \rightarrow 0$, and the overall $\iiint_V (\nabla^2 + k^2) G(\tilde{r}) dV \rightarrow 1$. So, we have shown that

$$\begin{cases} (\nabla^2 + k^2) G(\tilde{r}) = 0 & \text{for } \tilde{r} \neq 0 \\ \iiint_V (\nabla^2 + k^2) G(\tilde{r}) dV = 1 & \text{for } \tilde{r} = 0 \end{cases} \quad (10.8)$$

thus Green's function $G(\tilde{r}) = -\frac{1}{4\pi} \frac{e^{ik\tilde{r}}}{\tilde{r}}$ satisfies the equation (10.4).

Once we know the Green's function for the Helmholtz operator, we can find the general solution to Helmholtz equation (10.3). This is

$$\Xi(\mathbf{r}, s) = - \iiint_{V'} F(\mathbf{r}', s) G(\mathbf{r}, \mathbf{r}') dV' = \iiint_{V'} F(\mathbf{r}', s) \frac{e^{ik|\mathbf{r} - \mathbf{r}'|}}{4\pi|\mathbf{r} - \mathbf{r}'|} dV'. \quad (10.9)$$

Let us recall now that $k = i\frac{s}{c}$, so our s-domain potential function takes the following form

$$\Xi(\mathbf{r}, s) = \iiint_{V'} F(\mathbf{r}', s) \frac{e^{-s \frac{|\mathbf{r} - \mathbf{r}'|}{c}}}{4\pi|\mathbf{r} - \mathbf{r}'|} dV'. \quad (10.10)$$

To find potential $\xi(\mathbf{r}, t)$ in time domain, all we have to do now is to apply inverse Laplace transform

$$\xi(\mathbf{r}, t) = \mathcal{L}^{-1} \{ \Xi(\mathbf{r}, s) \}. \quad (10.11)$$

At this moment we will use yet another property of the Laplace transform. If $f(t)$ is the signal in time domain and $F(s)$ is its Laplace transform, then Laplace transform of the delayed signal is

$$\mathcal{L} \{ f(t - \tau) \} = F(s) e^{-s\tau}. \quad (10.12)$$

Consistently

$$f(t - \tau) = \mathcal{L}^{-1} \{ F(s) e^{-s\tau} \}. \quad (10.13)$$

Applying (10.13) to (10.10) we finally get

$$\xi(\mathbf{r}, t) = \frac{1}{4\pi} \iiint_{V'} \frac{f\left(\mathbf{r}', t - \frac{|\mathbf{r} - \mathbf{r}'|}{c}\right)}{|\mathbf{r} - \mathbf{r}'|} dV'. \quad (10.14)$$

This is it. Equation (10.14) is a general solution of the inhomogeneous three-dimensional wave equation (10.1).

11 Acknowledgments

This paper was brought to you by letters E, B, S and by three digits: 3, 5 and 7 (<https://radio357.pl>).

References

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